Advanced Solid State Physics (*P451*) Project Report

# The Parton Route to Quantum Spin Liquids: A General Overview

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I would like to thank Prof. Anamitra Mukherjee for teaching one of the best courses I have attended during my time at NISER so far. His lectures have rekindled my interest in pursuing research in condensed matter physics. I would also like to thank my group mates, Rabi and Uday, for their contributions and discussions throughout the project. We broadly divided the project into three parts. My report focuses mainly on the general approach and motivation behind parton mean-field theory. Rabi's report discusses the richer theoretical structure of the Kitaev model, including lattice gauge theory and symmetries. Uday's report focuses on the computational and numerical aspects of the project. All three reports can be found here [1].

## Disclaimer

I have tried to write this report in the style of David Tong's famous notes series, *Lectures on Theoretical Physics*; so, if I fail to create that impression, please note that I am simply not as smart as him (hopefully someday). At times, I have also used OpenAI's ChatGPT (Model: GPT-5.5, April 23, 2026 release) to paraphrase certain parts of the text. However, no new scientific content was generated through this use, and the final writing, organization, and interpretation are my own. One cartoon-style illustrative figure in the report was also generated using the same model. While care has been taken to preserve the meaning and technical accuracy of the content, readers are advised to be careful and, whenever needed, refer to the original cited resources. If you find any mistakes or inconsistencies, please contact me at `yash.chauhan@niser.ac.in`.

## Recommended Books and Resources

Here are a few general and broad resources that I have used at times while creating this report. The more specific references have been cited in the main text.

- Xiao-Gang Wen, *Quantum Field Theory of Many-body Systems*

  A great book, written very precisely. Although I have only read a few chapters, the background on parton mean-field theory was mainly studied from here.

- Subir Sachdev, *Quantum Phases of Matter*

  One of the best books I have come across in this area. Again, I have only read a small portion of it, but together with Prof. Sachdev's lectures on YouTube, it is a goldmine.

- Matthias Vojta, *Theory of Frustrated Magnetism*

  These are lecture notes by Prof. Vojta for a course at TU Dresden. They provide a nice refresher on magnetism, spin-wave theory, and quantum spin liquids.

- Jonas Sonnenschein, *Mean-field Theory and Projective Symmetry Group Classifications of Quantum Spin Liquids*

  This is a PhD thesis submitted at the Dahlem Center for Complex Quantum Systems. It has been a very helpful resource for studying parton mean-field theory, especially for anyone beginning to understand the technique.

## Map of the Report

This report follows a gradual path from ordinary magnetic order to quantum spin liquids, and then to the Kitaev–Heisenberg model. We begin by discussing magnetic order, frustration, and why a quantum spin liquid needs a language beyond classical spin patterns and magnons. The parton mean-field approach is then introduced using the Heisenberg model: spins are rewritten in terms of fermionic spinons, the physical constraint is imposed, and the exchange interaction is reorganized into hopping and pairing channels. After this, the Kitaev model is studied as the cleanest benchmark, where the same idea becomes exact through Majorana partons, itinerant matter Majoranas, and $\mathbb{Z}_2$ gauge fluxes. The main part of the report then turns to the Kitaev–Heisenberg model, where the Kitaev spin liquid competes with ordinary Heisenberg exchange. Here we introduce the effective Hamiltonian, discuss the parton mean-field viewpoint, use the four-sublattice rotation to understand the stripy antiferromagnetic phase as a hidden ferromagnet, and finally reproduce the main finite-cluster signatures of the phase diagram numerically. The report ends by connecting this theoretical story to experimental platforms such as $\alpha$-$RuCl_3$ (a section I wanted to include because my own interests are experimental), where real materials do not realize the pure Kitaev model exactly but still show signatures suggesting proximity to Kitaev spin-liquid physics.

# 1 The Basics

## 1.1 Introduction: From Magnetic Order to Spin Liquids

Magnets are familiar because, more often than not, they reveal themselves through order. In a ferromagnet, the spins point in the same direction; in an antiferromagnet, neighbouring spins point oppositely; and in more complicated magnets, the spins may twist into spirals, stripes, or larger repeating patterns. The details can vary a lot, but the basic idea is the same: at low temperature, the system chooses a pattern. This is what makes ordinary magnets relatively friendly to us. Once the magnetic pattern is known, the remaining quantum problem can often be treated as small fluctuations around that pattern. These fluctuations are the spin waves, or magnons, of the ordered magnet. In this sense, an ordered magnet gives us a classical picture first and then asks us to quantize the deviations from it.

Quantum mechanics, however, is not always so generous. In some magnetic systems, different local preferences compete so strongly that no single spin pattern can satisfy them all. A spin may want to be antiparallel to one neighbour, but the geometry of the lattice may make it impossible to be antiparallel to every neighbour at the same time. This is the essence of frustration: the system is asked to satisfy too many incompatible demands.

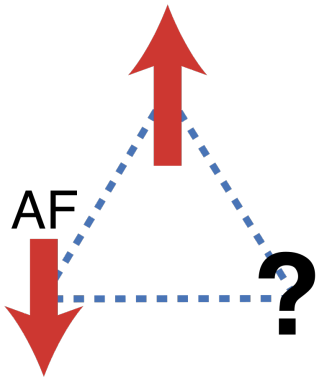



**Figure 1**: Magnetic frustration on a triangular unit. For antiferromagnetic interactions, each pair of neighbouring spins would prefer to align oppositely. On a triangle, satisfying two such bonds leaves the third bond unsatisfied, showing how lattice geometry can make the local interaction preferences mutually incompatible. Image adapted from Wikimedia Commons.

Frustration alone does not automatically produce a quantum spin liquid, since classical frustrated magnets can still order in nontrivial ways. The more interesting situation occurs when frustration is combined with strong quantum fluctuations. Then the spins may fail to settle into any static pattern and instead keep fluctuating among many competing possibilities.

The resulting state is not just a messy paramagnet. A hot paramagnet is disordered because temperature shakes the spins around. A quantum spin liquid is different: it remains disordered even at zero temperature, where thermal agitation has disappeared. Its lack of order is therefore not a sign that nothing interesting is happening, but rather that the organization is more subtle than ordinary magnetic order can capture. This is also why the usual language of magnons becomes insufficient. In a conventional magnet, magnons make sense because we expand around a known ordered state. In a quantum spin liquid, there is no obvious classical spin configuration to expand around; the system does not hand us a simple background of arrows.

The parton approach is one way of building a language for such phases. The idea is to rewrite each spin in terms of more elementary auxiliary degrees of freedom. This is not a free lunch: the new description contains extra states, so constraints must be imposed, and the rewriting introduces gauge redundancies. But precisely because of this redundancy, the parton language naturally brings in spinons, gauge fields, and mean-field descriptions of spin-liquid states. In this report, we follow this route step by step: first through the Heisenberg model, then through the exactly solvable Kitaev model, and finally toward the Kitaev–Heisenberg model, where spin-liquid physics competes with conventional magnetic order.

### 1.2 A Primer on Quantum Spin Liquids

A quantum spin liquid is a magnetic phase in which the spins remain strongly correlated, but do not form any static magnetic order even at zero temperature. This makes it very different from an ordinary paramagnet. A paramagnet is disordered because of thermal fluctuations, while a quantum spin liquid is disordered because of quantum fluctuations in the ground state itself. In a loose but useful sense, the ground state keeps moving in Hilbert space even when temperature has nothing left to do.

The usual starting point is a frustrated spin system, for example a spin Hamiltonian of the form

$$H = \sum_{\langle ij \rangle} J_{ij}\, \mathbf{S}_i \cdot \mathbf{S}_j. \tag{1.1}$$

For antiferromagnetic exchange, $J_{ij} > 0$, neighbouring spins prefer to be antiparallel. On certain lattices, however, this preference cannot be satisfied everywhere at once. The system is then forced to fluctuate among many nearly competing configurations instead of choosing one simple ordered pattern.

A useful picture is the resonating valence-bond idea. Two spin-$\frac{1}{2}$ moments can form a singlet,

$$|s\rangle_{ij} = \frac{1}{\sqrt{2}} \left( |\uparrow_i \downarrow_j\rangle - |\downarrow_i \uparrow_j\rangle \right). \tag{1.2}$$

A valence-bond solid chooses a fixed pattern of such singlets. A quantum spin liquid, roughly speaking, is closer to a quantum superposition of many such singlet patterns,

$$|\Psi_{\mathrm{RVB}}\rangle = \sum_{\mathcal{C}} A_{\mathcal{C}} \prod_{(ij)\in\mathcal{C}} |s\rangle_{ij}\,, \tag{1.3}$$

where $\mathcal{C}$ labels a particular covering of the lattice by singlet bonds and $A_{\mathcal{C}}$ is its quantum amplitude. The spins are not independent, but the correlations do not freeze into a conventional magnetic order.

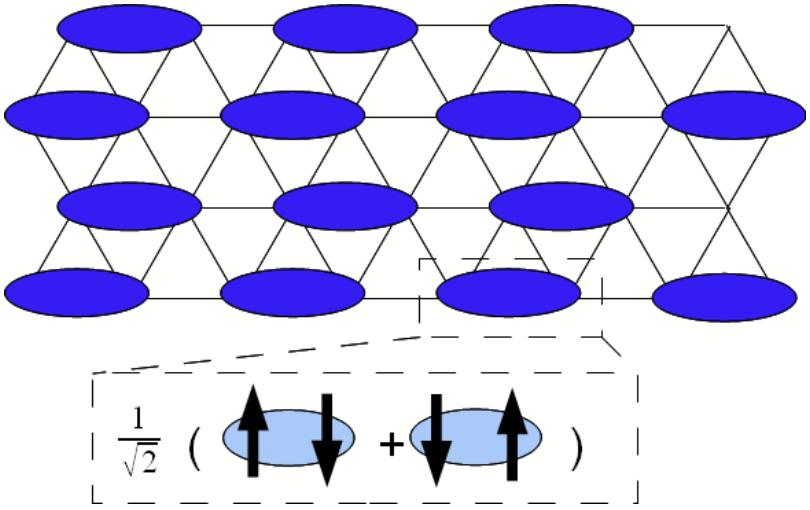



**Figure 2**: Valence bonds and the RVB idea. A fixed arrangement of singlet bonds gives a valence-bond solid, while an RVB-like spin liquid is better viewed as a quantum superposition of many different singlet coverings. The precise singlet convention used in this report is the antisymmetric combination written in Eq. (1.2); the figure is used only as a schematic picture of bond coverings and resonance. Image adapted from Wikimedia Commons.

The important ingredients of a quantum spin liquid are:

- **Absence of magnetic long-range order:** The local magnetic moment does not develop a static expectation value,

$$\langle \mathbf{S}_i \rangle = 0. \tag{1.4}$$

  Equivalently, there is no conventional magnetic order parameter. In many cases the spin correlations also fail to approach a nonzero constant at large separation,

$$\lim_{|i-j|\to\infty} \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle = 0. \tag{1.5}$$

  However, this alone is not enough to define a spin liquid, since a trivial paramagnet can also satisfy this condition.

- **Strong quantum fluctuations:** The spins continue to fluctuate even at zero temperature. These fluctuations are not thermal, but are part of the quantum ground state.

- **Frustration:** Competing interactions or lattice geometry prevent the system from satisfying all local spin preferences simultaneously. This helps suppress simple magnetic order.

- **Long-range entanglement:** The ground state cannot be understood as a simple product state or a slightly disordered classical magnet. Its structure is genuinely quantum mechanical and non-local.

- **Fractionalized excitations:** The elementary excitations need not be ordinary magnons. In many spin liquids, a spin excitation can split into spin-$\frac{1}{2}$ quasiparticles called spinons. Schematically, one may think of this as

$$S = 1 \quad \longrightarrow \quad S = \frac{1}{2} + S = \frac{1}{2}. \tag{1.6}$$

- **Emergent gauge structure:** The low-energy theory often contains gauge fields that are not present explicitly in the microscopic Hamiltonian. These gauge fields arise from the redundancy used to describe fractionalized degrees of freedom.

Thus, a quantum spin liquid is not simply a phase with no order. It is a phase where the usual language of magnetic order and magnons becomes insufficient. Instead, one needs a framework that can naturally describe fractionalized particles, gauge redundancy, and highly entangled spin states. This is precisely the motivation for using the parton approach.

### 1.3 Why Partons?

The main difficulty with a quantum spin liquid is that there is no simple ordered background to expand around. In an ordered magnet, the low-energy excitations can often be described as spin waves above a classical spin pattern. For a spin liquid, this is not the natural starting point, because the ground state is already a strongly entangled superposition of many spin configurations. The parton approach takes a different route: instead of expanding around an ordered state, we rewrite each spin in terms of auxiliary particles.

A common choice is the fermionic, or slave-fermion, representation,

$$S_i^\mu = \frac{1}{2} f_{i\alpha}^\dagger \sigma_{\alpha\beta}^\mu f_{i\beta}, \qquad \mu = x, y, z. \tag{1.7}$$

Here $f_{i\alpha}$ is an auxiliary fermion carrying spin index $\alpha = \uparrow, \downarrow$, and $\sigma^\mu$ are the Pauli matrices. If these fermions survive as low-energy excitations after all constraints are properly treated, they are called spinons.

At first sight, this rewriting seems to make the problem worse. A spin-$\frac{1}{2}$ has only two local states, $|\uparrow\rangle$ and $|\downarrow\rangle$. The fermionic description has four: empty, spin-up, spin-down, and doubly occupied. Only the singly occupied states are physical, so we must impose the local constraint

$$\sum_\alpha f_{i\alpha}^\dagger f_{i\alpha} = 1. \tag{1.8}$$

This is the price of the parton trick: we gain a useful fermionic language, but we enlarge the Hilbert space and must later remove the unphysical states.

The usefulness of this rewriting is that a spin Hamiltonian becomes an interacting fermion problem, which can then be treated using mean-field theory. The resulting mean-field state may contain spinon hopping, spinon pairing, or gauge fluxes. These are not just

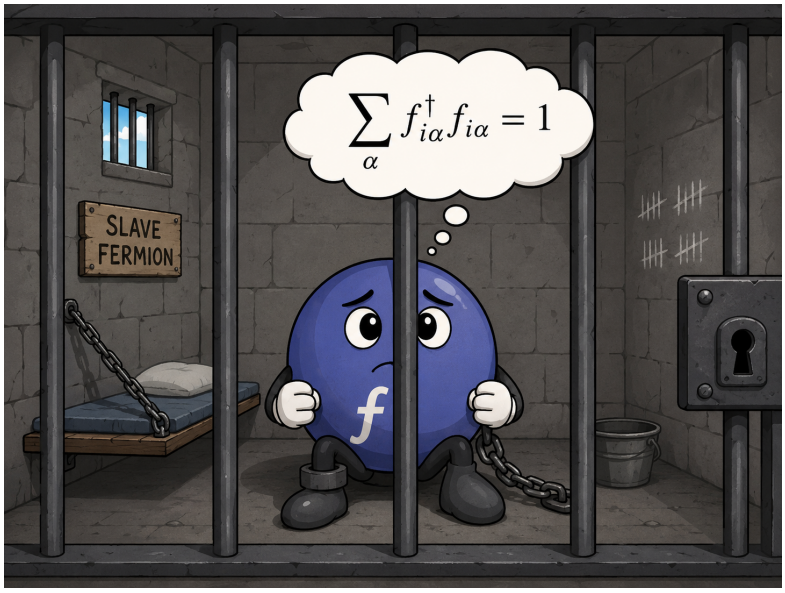



**Figure 3**: Cartoonish illustration of the slave-fermion representation. The fermionic description enlarges the local Hilbert space from the two physical spin states to four fermionic states. The single-occupancy constraint in Eq. (1.8) removes the empty and doubly occupied states, leaving only the physical spin-$\frac{1}{2}$ subspace. Image generated by ChatGPT.

mathematical decorations; they are the natural structures one expects in fractionalized spin-liquid phases. This viewpoint grew out of Anderson's resonating valence-bond idea, where a spin liquid is imagined as a quantum superposition of many singlet-bond coverings rather than a frozen pattern of magnetic moments.

The constraint also tells us something deeper. The parton representation has a redundancy: different parton states can describe the same physical spin state. For example,

$$f_{i\alpha} \to e^{i\theta_i} f_{i\alpha} \tag{1.9}$$

leaves the spin operator in Eq. (1.7) unchanged. This local redundancy is the origin of emergent gauge fields in the low-energy theory. Thus the parton approach does two things at once: it makes fractionalized excitations visible, and it naturally leads to gauge structure. This is why it is such a useful language for quantum spin liquids, and why it provides a natural starting point for studying the Kitaev and Kitaev–Heisenberg models.

### 1.4 Mean-Field Theory and Its Limitations

The parton construction gives us a new language, but it does not solve the spin problem by itself. After writing the spin operators in terms of fermionic spinons, a simple-looking spin Hamiltonian usually becomes quartic in the fermion operators. For example, the Heisenberg interaction can be written schematically as

$$\mathbf{S}_i \cdot \mathbf{S}_j \sim -\frac{1}{2}\left(f_{i\alpha}^\dagger f_{j\alpha}\right)\left(f_{j\beta}^\dagger f_{i\beta}\right) + \cdots, \tag{1.10}$$

where the dots contain density terms and constraint-related pieces. So the price of making spinons visible is that we now have an interacting fermion problem. Mean-field theory is the step where we replace this difficult many-body problem by an effective single-particle problem, while keeping the most important average correlations.

The basic mean-field trick is to split operators into their average value and fluctuation,

$$A = \langle A \rangle + \delta A, \qquad B = \langle B \rangle + \delta B. \tag{1.11}$$

Then

$$AB = \langle A \rangle \langle B \rangle + \langle A \rangle \delta B + \delta A \langle B \rangle + \delta A \delta B. \tag{1.12}$$

Mean-field theory neglects the quadratic fluctuation term $\delta A \delta B$, giving

$$AB \approx A\langle B \rangle + \langle A \rangle B - \langle A \rangle \langle B \rangle. \tag{1.13}$$

This is the useful simplification: the interacting Hamiltonian becomes quadratic and can usually be diagonalized.

In parton mean-field theory, the averages are chosen according to the kind of spin-liquid state we want to describe. Two common bond fields are the spinon hopping and pairing amplitudes,

$$\chi_{ij} = \sum_{\alpha} \left\langle f_{i\alpha}^{\dagger} f_{j\alpha} \right\rangle, \qquad \Delta_{ij} = \langle f_{i\uparrow} f_{j\downarrow} - f_{i\downarrow} f_{j\uparrow} \rangle \,. \tag{1.14}$$

Here $\chi_{ij}$ describes spinon hopping, while $\Delta_{ij}$ describes spinon pairing. Different patterns of $\chi_{ij}$, $\Delta_{ij}$, and possible magnetic order parameters correspond to different candidate phases. The single-occupancy constraint is also usually imposed only on average,

$$H_{\text{constraint}} = \sum_{i} \lambda_i \left( \sum_{\alpha} f_{i\alpha}^{\dagger} f_{i\alpha} - 1 \right), \tag{1.15}$$

so that the schematic mean-field Hamiltonian becomes

$$H_{\text{MF}} = H_{\text{spinon}}(\chi, \Delta) + H_{\text{constraint}}. \tag{1.16}$$

At this level, the spin liquid appears as a band theory or pairing theory of spinons.
**This is also where the main danger enters.** The mean-field state is not automatically a physical spin state. First, the constraint has usually been imposed only on average, so the wavefunction may still contain unphysical empty and doubly occupied configurations before projection. Second, the parton representation has gauge redundancy: many different parton configurations describe the same physical spin configuration. Mean-field theory freezes the corresponding gauge fields at a saddle point, but gauge fluctuations can be decisive. They may confine spinons, change the low-energy spectrum, or even destroy a proposed spin-liquid phase. Third, the answer depends on the chosen ansatz; a mean-field calculation only explores the hopping, pairing, and magnetic channels we allow it to explore.

Thus, parton mean-field theory should not be read as an exact solution of a spin model. It is better viewed as a way of constructing and organizing possible spin-liquid phases. When gauge fluctuations are weak or gapped, as in stable $\mathbb{Z}_2$ spin liquids, the mean-field picture can be physically meaningful. When gauge fluctuations are strong, the mean-field state may be only a useful starting point. In this report, we will use this attitude throughout: first we will see how the parton machinery works for the Heisenberg model, then use the Kitaev model as a rare case where fractionalization is exact, and finally study the Kitaev–Heisenberg model where the spin-liquid tendency has to compete with ordinary magnetic order.

This caveat is especially important for the Kitaev–Heisenberg problem. Near the pure Kitaev limit, the fractionalized description is strongly motivated by the exact Kitaev solution. Away from this limit, additional interactions make the problem less protected: gauge fields become dynamical, spinons may no longer remain deconfined, and magnetic order can compete with the spin-liquid state. The value of the parton mean-field approach is therefore not that it magically solves the full problem, but that it gives a systematic way to follow this competition between fractionalized spin-liquid physics and conventional magnetic order.

## 2 Parton Approach to the Heisenberg Model

### 2.1 The Heisenberg Model as the Starting Point

The simplest place to begin is the spin-$\frac{1}{2}$ Heisenberg model,

$$H = \sum_{\langle ij \rangle} J_{ij}\, \mathbf{S}_i \cdot \mathbf{S}_j. \tag{2.1}$$

Here $\mathbf{S}_i$ is a spin-$\frac{1}{2}$ operator on site $i$, and $J_{ij}$ is the exchange coupling between neighbouring sites. For $J_{ij} < 0$, the spins prefer to align parallel, giving ferromagnetic tendencies. For $J_{ij} > 0$, the spins prefer to be antiparallel, giving antiferromagnetic tendencies. This model is simple to write down, but it already contains the main tension behind quantum magnetism: each spin wants to satisfy its bonds with its neighbours, but the lattice may not allow all these local wishes to be satisfied at once.

To see why the antiferromagnetic case is naturally connected to valence bonds, consider just two spins with Hamiltonian

$$H_{ij} = J\, \mathbf{S}_i \cdot \mathbf{S}_j, \qquad J > 0. \tag{2.2}$$

Using

$$\mathbf{S}_i \cdot \mathbf{S}_j = \frac{1}{2}\left[(\mathbf{S}_i + \mathbf{S}_j)^2 - \mathbf{S}_i^2 - \mathbf{S}_j^2\right], \tag{2.3}$$

the two-spin problem separates into a singlet and a triplet sector. The singlet has total spin $S_{\text{tot}} = 0$ and energy

$$E_s = -\frac{3J}{4}, \tag{2.4}$$

while the triplet has total spin $S_{\text{tot}} = 1$ and energy

$$E_t = \frac{J}{4}. \tag{2.5}$$

So an antiferromagnetic bond likes to form a spin singlet. This is the microscopic reason why singlets and valence bonds appear so naturally in discussions of quantum spin liquids.

On a full lattice, however, the situation is no longer just a collection of independent two-spin problems. One spin belongs to several bonds at once, and it cannot form a perfect singlet with every neighbour simultaneously. On unfrustrated bipartite lattices, the system often resolves this competition by developing Néel order. Then the spin-wave picture is useful. But on frustrated lattices, or in systems with competing interactions, the singlet tendency may fail to freeze into one simple pattern. Instead, many different valence-bond coverings can become important.

This is why the Heisenberg model is a natural starting point for parton mean-field theory. It is simple enough that the algebra can be shown explicitly, but rich enough to contain the essential question: should the system form conventional magnetic order, or can it remain in a highly entangled, non-magnetic state? In the next steps, we will rewrite the spin operators in terms of fermionic spinons, see how the exchange interaction becomes a four-fermion problem, and then use mean-field decoupling to turn that problem into candidate spin-liquid ansätze.

### 2.2 Fermionic Spinons and the Physical Constraint

The parton approach begins by rewriting each spin in terms of auxiliary fermions. For a spin-$\frac{1}{2}$ system, we introduce two fermionic operators on every site,

$$f_{i\uparrow}, \qquad f_{i\downarrow},$$

where the index $\uparrow,\downarrow$ labels the spin carried by the auxiliary fermion. These operators satisfy the usual fermionic anticommutation relations,

$$\{f_{i\alpha}, f_{j\beta}^\dagger\} = \delta_{ij}\delta_{\alpha\beta}, \qquad \{f_{i\alpha}, f_{j\beta}\} = \{f_{i\alpha}^\dagger, f_{j\beta}^\dagger\} = 0. \tag{2.6}$$

The spin operator is then represented as

$$S_i^\mu = \frac{1}{2} f_{i\alpha}^\dagger \sigma_{\alpha\beta}^\mu f_{i\beta}, \qquad \mu = x, y, z, \tag{2.7}$$

where $\sigma^\mu$ are the Pauli matrices and repeated spin indices are summed over. In components this gives

$$S_i^z = \frac{1}{2}\left(f_{i\uparrow}^\dagger f_{i\uparrow} - f_{i\downarrow}^\dagger f_{i\downarrow}\right), \tag{2.8}$$

$$S_i^+ = f_{i\uparrow}^\dagger f_{i\downarrow}, \tag{2.9}$$

$$S_i^- = f_{i\downarrow}^\dagger f_{i\uparrow}. \tag{2.10}$$

This makes the interpretation simple: $S_i^+$ converts a down spinon into an up spinon, while $S_i^-$ converts an up spinon into a down spinon.

The catch is that the fermionic description is too large. A single spin-$\frac{1}{2}$ has only two local states,

$$|\uparrow\rangle_i, \qquad |\downarrow\rangle_i. \tag{2.11}$$

But the fermionic Hilbert space on one site has four states:

$$|0\rangle_i, \qquad f_{i\uparrow}^\dagger|0\rangle_i, \qquad f_{i\downarrow}^\dagger|0\rangle_i, \qquad f_{i\uparrow}^\dagger f_{i\downarrow}^\dagger|0\rangle_i. \tag{2.12}$$

Only the singly occupied states are physical:

$$|\uparrow\rangle_i \equiv f_{i\uparrow}^\dagger|0\rangle_i, \qquad |\downarrow\rangle_i \equiv f_{i\downarrow}^\dagger|0\rangle_i. \tag{2.13}$$

The empty state and the doubly occupied state are artifacts of the rewriting and must be removed.

This is done by imposing the single-occupancy constraint

$$n_i = \sum_\alpha f_{i\alpha}^\dagger f_{i\alpha} = 1. \tag{2.14}$$

Physically, this says that every site must contain exactly one spinon. After this projection, the fermionic Hilbert space collapses back to the original two-dimensional spin-$\frac{1}{2}$ Hilbert space. One may also write the related pair constraints

$$\epsilon_{\alpha\beta} f_{i\alpha} f_{i\beta} = 0, \qquad \epsilon_{\alpha\beta} f_{i\alpha}^\dagger f_{i\beta}^\dagger = 0, \tag{2.15}$$

which remove the unphysical empty/doubly occupied sector in an equivalent way once the physical subspace is imposed.

The important point is that spinons are not ordinary electrons freely moving through the lattice. They are auxiliary particles introduced to rewrite the spin problem. Only after imposing the constraint does the fermionic description become equivalent to the original spin model. At the same time, this enlargement is exactly what makes the method useful: once spins are written as fermions, the exchange Hamiltonian becomes an interacting spinon Hamiltonian, and we can start discussing hopping, pairing, and mean-field spin-liquid ansätze.

### 2.3 Rewriting the Exchange Interaction

We now rewrite the Heisenberg exchange interaction in the fermionic spinon language. The object we want to understand is

$$\mathbf{S}_i \cdot \mathbf{S}_j = \sum_{\mu=x,y,z} S_i^\mu S_j^\mu. \tag{2.16}$$

Using

$$S_i^\mu = \frac{1}{2} f_{i\alpha}^\dagger \sigma_{\alpha\beta}^\mu f_{i\beta}, \tag{2.17}$$

we get

$$\begin{aligned} \mathbf{S}_i \cdot \mathbf{S}_j &= \sum_\mu \left( \frac{1}{2} f_{i\alpha}^\dagger \sigma_{\alpha\beta}^\mu f_{i\beta} \right) \left( \frac{1}{2} f_{j\gamma}^\dagger \sigma_{\gamma\delta}^\mu f_{j\delta} \right) \\ &= \frac{1}{4} f_{i\alpha}^\dagger f_{i\beta} f_{j\gamma}^\dagger f_{j\delta} \sum_\mu \sigma_{\alpha\beta}^\mu \sigma_{\gamma\delta}^\mu. \end{aligned} \tag{2.18}$$

Here and below, repeated spin indices are summed over. The Pauli matrices satisfy

$$\sum_{\mu=x,y,z} \sigma_{\alpha\beta}^\mu \sigma_{\gamma\delta}^\mu = 2\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\gamma\delta}. \tag{2.19}$$

Therefore,

$$\begin{aligned} \mathbf{S}_i \cdot \mathbf{S}_j &= \frac{1}{4} f_{i\alpha}^\dagger f_{i\beta} f_{j\gamma}^\dagger f_{j\delta} \left( 2\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\gamma\delta} \right) \\ &= \frac{1}{2} f_{i\alpha}^\dagger f_{i\beta} f_{j\beta}^\dagger f_{j\alpha} - \frac{1}{4} f_{i\alpha}^\dagger f_{i\alpha} f_{j\gamma}^\dagger f_{j\gamma}. \end{aligned} \tag{2.20}$$

Introducing the spinon number operator

$$n_i = \sum_\alpha f_{i\alpha}^\dagger f_{i\alpha}, \tag{2.21}$$

this becomes

$$\mathbf{S}_i \cdot \mathbf{S}_j = \frac{1}{2} f_{i\alpha}^\dagger f_{i\beta} f_{j\beta}^\dagger f_{j\alpha} - \frac{1}{4} n_i n_j. \tag{2.22}$$

For $i \neq j$, fermionic operators on different sites anticommute. Thus

$$f_{i\beta} f_{j\beta}^\dagger = -f_{j\beta}^\dagger f_{i\beta}, \tag{2.23}$$

and the first term in Eq. (2.22) can be rewritten as

$$f^\dagger_{i\alpha} f_{i\beta} f^\dagger_{j\beta} f_{j\alpha} = -f^\dagger_{i\alpha} f^\dagger_{j\beta} f_{i\beta} f_{j\alpha}. \tag{2.24}$$

This is already the important structural result: the original two-spin exchange has become a four-fermion interaction.

It is useful to package this four-fermion term into bond operators. The first natural choice is the particle–hole, or hopping, channel:

$$\hat{\chi}_{ij} = \sum_\alpha f^\dagger_{i\alpha} f_{j\alpha}, \qquad \hat{\chi}_{ji} = \sum_\alpha f^\dagger_{j\alpha} f_{i\alpha}. \tag{2.25}$$

Then

$$\begin{aligned} \hat{\chi}_{ij}\hat{\chi}_{ji} &= f^\dagger_{i\alpha} f_{j\alpha} f^\dagger_{j\beta} f_{i\beta} \\ &= f^\dagger_{i\alpha} \left(\delta_{\alpha\beta} - f^\dagger_{j\beta} f_{j\alpha}\right) f_{i\beta} \\ &= n_i - f^\dagger_{i\alpha} f^\dagger_{j\beta} f_{j\alpha} f_{i\beta}. \end{aligned} \tag{2.26}$$

Using $f_{j\alpha} f_{i\beta} = -f_{i\beta} f_{j\alpha}$, this becomes

$$\hat{\chi}_{ij}\hat{\chi}_{ji} = n_i + f^\dagger_{i\alpha} f^\dagger_{j\beta} f_{i\beta} f_{j\alpha}. \tag{2.27}$$

Hence

$$f^\dagger_{i\alpha} f^\dagger_{j\beta} f_{i\beta} f_{j\alpha} = \hat{\chi}_{ij}\hat{\chi}_{ji} - n_i. \tag{2.28}$$

Substituting this into Eq. (2.22), we find

$$\mathbf{S}_i \cdot \mathbf{S}_j = -\frac{1}{2}\hat{\chi}_{ij}\hat{\chi}_{ji} + \frac{1}{2} n_i - \frac{1}{4} n_i n_j. \tag{2.29}$$

Inside the physical Hilbert space, the single-occupancy constraint gives

$$n_i = n_j = 1. \tag{2.30}$$

Therefore the number terms are constants, and we may write

$$\mathbf{S}_i \cdot \mathbf{S}_j = -\frac{1}{2}\hat{\chi}_{ij}\hat{\chi}_{ji} + \frac{1}{4}, \qquad \text{in the physical subspace.} \tag{2.31}$$

Equivalently, up to a constant fixed by the constraint,

$$\mathbf{S}_i \cdot \mathbf{S}_j \sim -\frac{1}{2}\hat{\chi}_{ij}\hat{\chi}_{ji}. \tag{2.32}$$

The Heisenberg Hamiltonian can therefore be written, up to constants, as

$$\begin{aligned} H &= \sum_{\langle ij \rangle} J_{ij}\, \mathbf{S}_i \cdot \mathbf{S}_j \\ &\sim -\frac{1}{2} \sum_{\langle ij \rangle} J_{ij}\, \hat{\chi}_{ij}\hat{\chi}_{ji} \\ &= -\frac{1}{2} \sum_{\langle ij \rangle} J_{ij} \left(f^\dagger_{i\alpha} f_{j\alpha}\right) \left(f^\dagger_{j\beta} f_{i\beta}\right). \end{aligned} \tag{2.33}$$

This form suggests a hopping mean-field ansatz: the spinons acquire an average amplitude to move from one site to another.

There is a second, equally important way of writing the same antiferromagnetic exchange. Define the particle–particle singlet pairing operator on the bond $\langle ij \rangle$ by

$$\hat{\Delta}_{ij}^{\dagger} = f_{i\uparrow}^{\dagger} f_{j\downarrow}^{\dagger} - f_{i\downarrow}^{\dagger} f_{j\uparrow}^{\dagger}, \tag{2.34}$$

with

$$\hat{\Delta}_{ij} = f_{j\downarrow} f_{i\uparrow} - f_{j\uparrow} f_{i\downarrow}. \tag{2.35}$$

The operator $\hat{\Delta}_{ij}^{\dagger}$ creates a spin singlet on the bond. In the physical subspace one obtains the equivalent identity

$$\mathbf{S}_i \cdot \mathbf{S}_j = -\frac{1}{2} \hat{\Delta}_{ij}^{\dagger} \hat{\Delta}_{ij} + \frac{1}{4}. \tag{2.36}$$

This version makes the antiferromagnetic physics especially transparent: for $J_{ij} > 0$, the energy is lowered when neighbouring spinons form singlet correlations.

Thus the same Heisenberg exchange admits two useful mean-field languages:

$$\text{particle–hole channel:} \qquad \hat{\chi}_{ij} = f_{i\alpha}^{\dagger} f_{j\alpha}, \tag{2.37}$$

and

$$\text{particle–particle channel:} \qquad \hat{\Delta}_{ij} = f_{j\downarrow} f_{i\uparrow} - f_{j\uparrow} f_{i\downarrow}. \tag{2.38}$$

The hopping channel describes spinons moving through the lattice, while the pairing channel describes resonating singlet bonds. Both are legitimate ways to organize the same interaction. In the next subsection, we will first use the hopping channel because it gives the cleanest route to a quadratic spinon Hamiltonian and makes the appearance of gauge phases particularly visible. The pairing channel will remain important in the background, especially when we later connect the mean-field language to RVB physics and spin-liquid ansätze.

### 2.4 Mean-Field Decoupling and the Spinon Hamiltonian

We have now reached the practical point of the parton construction. The Heisenberg exchange has been rewritten as a quartic interaction between spinons. In the particle–hole, or hopping, channel we found

$$H \simeq -\frac{1}{2} \sum_{\langle ij \rangle} J_{ij} \, \hat{\chi}_{ij} \hat{\chi}_{ji}, \tag{2.39}$$

where

$$\hat{\chi}_{ij} = \sum_{\alpha} f_{i\alpha}^{\dagger} f_{j\alpha}, \qquad \hat{\chi}_{ji} = \sum_{\alpha} f_{j\alpha}^{\dagger} f_{i\alpha}. \tag{2.40}$$

The symbol $\simeq$ reminds us that constant terms fixed by the single-occupancy constraint have been dropped. The problem is now written in a useful language, but it is still not solved: Eq. (2.39) is still an interacting many-body Hamiltonian.

The mean-field step is to replace the bond operator by its expectation value plus a fluctuation,

$$\hat{\chi}_{ij} = \chi_{ij} + \delta\hat{\chi}_{ij}, \qquad \chi_{ij} = \langle\hat{\chi}_{ij}\rangle = \sum_{\alpha} \left\langle f^{\dagger}_{i\alpha} f_{j\alpha} \right\rangle . \tag{2.41}$$

Similarly,

$$\hat{\chi}_{ji} = \chi_{ji} + \delta\hat{\chi}_{ji}, \qquad \chi_{ji} = \langle\hat{\chi}_{ji}\rangle. \tag{2.42}$$

Since $\hat{\chi}_{ji} = \hat{\chi}^{\dagger}_{ij}$, we usually have

$$\chi_{ji} = \chi^{*}_{ij}. \tag{2.43}$$

Now expand the product:

$$\begin{aligned} \hat{\chi}_{ij}\hat{\chi}_{ji} &= \left(\chi_{ij} + \delta\hat{\chi}_{ij}\right)\left(\chi_{ji} + \delta\hat{\chi}_{ji}\right) \\ &= \chi_{ij}\chi_{ji} + \chi_{ij}\delta\hat{\chi}_{ji} + \delta\hat{\chi}_{ij}\chi_{ji} + \delta\hat{\chi}_{ij}\delta\hat{\chi}_{ji}. \end{aligned} \tag{2.44}$$

Mean-field theory neglects the term quadratic in fluctuations,

$$\delta\hat{\chi}_{ij}\delta\hat{\chi}_{ji} \approx 0. \tag{2.45}$$

Thus

$$\begin{aligned} \hat{\chi}_{ij}\hat{\chi}_{ji} &\approx \chi_{ij}\chi_{ji} + \chi_{ij}\left(\hat{\chi}_{ji} - \chi_{ji}\right) + \left(\hat{\chi}_{ij} - \chi_{ij}\right)\chi_{ji} \\ &= \chi_{ij}\hat{\chi}_{ji} + \chi_{ji}\hat{\chi}_{ij} - \chi_{ij}\chi_{ji}. \end{aligned} \tag{2.46}$$

Using $\chi_{ji} = \chi^{*}_{ij}$, this becomes

$$\hat{\chi}_{ij}\hat{\chi}_{ji} \approx \chi_{ij}\hat{\chi}_{ji} + \chi^{*}_{ij}\hat{\chi}_{ij} - |\chi_{ij}|^2. \tag{2.47}$$

Substituting this into Eq. (2.39), we obtain

$$\begin{aligned} H^{(\chi)}_{\mathrm{MF}} &= -\frac{1}{2}\sum_{\langle ij\rangle} J_{ij}\left(\chi_{ij}\hat{\chi}_{ji} + \chi^{*}_{ij}\hat{\chi}_{ij} - |\chi_{ij}|^2\right) \\ &= \frac{1}{2}\sum_{\langle ij\rangle} J_{ij}|\chi_{ij}|^2 - \frac{1}{2}\sum_{\langle ij\rangle,\alpha} J_{ij}\left(\chi_{ij} f^{\dagger}_{j\alpha} f_{i\alpha} + \chi^{*}_{ij} f^{\dagger}_{i\alpha} f_{j\alpha}\right). \end{aligned} \tag{2.48}$$

This is the key simplification: the Hamiltonian is now quadratic in the spinon operators. The original exchange interaction has become an effective hopping problem for spinons, and $\chi_{ij}$ plays the role of the hopping amplitude on the bond $\langle ij\rangle$.

The constraint must also be included. The exact physical condition is

$$\sum_{\alpha} f^{\dagger}_{i\alpha} f_{i\alpha} = 1 \qquad \text{on every site } i. \tag{2.49}$$

At mean-field level, this is imposed using a Lagrange multiplier $\lambda_i$:

$$H_{\lambda} = \sum_{i} \lambda_i \left(\sum_{\alpha} f^{\dagger}_{i\alpha} f_{i\alpha} - 1\right). \tag{2.50}$$

So the full hopping mean-field Hamiltonian is

$$H_{\mathrm{MF}}^{(\chi)} = \frac{1}{2}\sum_{\langle ij\rangle} J_{ij}|\chi_{ij}|^2 - \frac{1}{2}\sum_{\langle ij\rangle,\alpha} J_{ij}\left(\chi_{ij} f_{j\alpha}^\dagger f_{i\alpha} + \chi_{ij}^* f_{i\alpha}^\dagger f_{j\alpha}\right) + \sum_i \lambda_i \left(\sum_\alpha f_{i\alpha}^\dagger f_{i\alpha} - 1\right). \tag{2.51}$$

The parameters $\chi_{ij}$ and $\lambda_i$ are not arbitrary. They must be determined self-consistently from the ground state of the same Hamiltonian:

$$\chi_{ij} = \sum_\alpha \left\langle f_{i\alpha}^\dagger f_{j\alpha}\right\rangle_{\mathrm{MF}}, \tag{2.52}$$

and

$$\sum_\alpha \left\langle f_{i\alpha}^\dagger f_{i\alpha}\right\rangle_{\mathrm{MF}} = 1. \tag{2.53}$$

The first equation fixes the hopping amplitude, while the second fixes the Lagrange multiplier. Together they say that the mean field must be consistent with the state it produces.

For a translationally invariant ansatz, one often takes

$$\chi_{ij} = \chi, \qquad \lambda_i = \lambda, \tag{2.54}$$

possibly with signs or phases depending on the chosen flux pattern. Then the Hamiltonian can be diagonalized by Fourier transformation. For example, with one site per unit cell,

$$f_{i\alpha} = \frac{1}{\sqrt{N}}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}_i} f_{\mathbf{k}\alpha}, \tag{2.55}$$

and the Hamiltonian takes the schematic form

$$H_{\mathrm{MF}}^{(\chi)} = E_0 + \sum_{\mathbf{k},\alpha} \varepsilon(\mathbf{k})\, f_{\mathbf{k}\alpha}^\dagger f_{\mathbf{k}\alpha}. \tag{2.56}$$

Here $E_0$ contains the constant mean-field energy, and $\varepsilon(\mathbf{k})$ is the spinon dispersion. The precise form of $\varepsilon(\mathbf{k})$ depends on the lattice and on the pattern of $\chi_{ij}$. If $\chi_{ij}$ is real and uniform, the spinons hop with the same amplitude on every bond. If $\chi_{ij}$ is complex, the spinons effectively move in a background gauge flux.

There is also a second natural decoupling: the particle–particle, or singlet pairing, channel. Define

$$\hat{\Delta}_{ij} = f_{j\downarrow} f_{i\uparrow} - f_{j\uparrow} f_{i\downarrow}, \qquad \Delta_{ij} = \left\langle \hat{\Delta}_{ij}\right\rangle. \tag{2.57}$$

Using

$$\mathbf{S}_i \cdot \mathbf{S}_j = -\frac{1}{2}\hat{\Delta}_{ij}^\dagger \hat{\Delta}_{ij} + \frac{1}{4}, \tag{2.58}$$

the same mean-field logic gives

$$\hat{\Delta}_{ij}^\dagger \hat{\Delta}_{ij} \approx \Delta_{ij}^* \hat{\Delta}_{ij} + \Delta_{ij}\hat{\Delta}_{ij}^\dagger - |\Delta_{ij}|^2. \tag{2.59}$$

Therefore,

$$H_{\mathrm{MF}}^{(\Delta)} = \frac{1}{2}\sum_{\langle ij\rangle} J_{ij}|\Delta_{ij}|^2 - \frac{1}{2}\sum_{\langle ij\rangle} J_{ij}\left(\Delta_{ij}^{*}\hat{\Delta}_{ij} + \Delta_{ij}\hat{\Delta}_{ij}^{\dagger}\right). \tag{2.60}$$

This describes spinon pairing rather than spinon hopping, and it is the mean-field version of resonating singlet bonds.

Both channels are useful, but they emphasize different physics. The hopping channel makes spinon motion and gauge fluxes especially transparent, so we will mainly use it here to explain the basic structure of parton mean-field theory. The pairing channel is equally important in more general RVB descriptions, where the spin liquid is naturally viewed as a fluctuating sea of singlet bonds. In a full singlet parton ansatz, hopping and pairing may both appear, and the resulting Hamiltonian is a Bogoliubov–de Gennes-type spinon Hamiltonian rather than a simple hopping Hamiltonian.

The crucial lesson is that mean-field theory makes the spinons look like free particles, but they are not yet free physical particles. They still live in an enlarged Hilbert space, and the constraint has only been imposed on average. The fluctuations of $\chi_{ij}$, $\Delta_{ij}$, and $\lambda_i$ are therefore not optional details: they are tied to the gauge structure of the parton representation. This is the point we turn to next.

### 2.5 Gauge Redundancy and What We Learn

We now come to the most important warning in the parton construction. The fermionic spinons are useful, but they are not unique. The same physical spin operator can be represented by many different spinon configurations. This is because the spin operator

$$S_i^{\mu} = \frac{1}{2} f_{i\alpha}^{\dagger}\sigma_{\alpha\beta}^{\mu} f_{i\beta} \tag{2.61}$$

is unchanged under a local phase rotation

$$f_{i\alpha} \to e^{i\theta_i} f_{i\alpha}. \tag{2.62}$$

Indeed,

$$f_{i\alpha}^{\dagger}\sigma_{\alpha\beta}^{\mu} f_{i\beta} \to e^{-i\theta_i} f_{i\alpha}^{\dagger}\sigma_{\alpha\beta}^{\mu} e^{i\theta_i} f_{i\beta} = f_{i\alpha}^{\dagger}\sigma_{\alpha\beta}^{\mu} f_{i\beta}. \tag{2.63}$$

So the phase of the spinon is not directly physical. It is a redundancy of the description, not a symmetry acting on the original spin system. This is the origin of the emergent gauge structure.

The mean-field hopping parameter transforms under the same local phase rotation. Since

$$\chi_{ij} = \sum_{\alpha}\left\langle f_{i\alpha}^{\dagger} f_{j\alpha}\right\rangle, \tag{2.64}$$

we get

$$\chi_{ij} \to e^{-i\theta_i}\chi_{ij} e^{i\theta_j}. \tag{2.65}$$

Thus the phase of a single $\chi_{ij}$ is not gauge invariant. It can be changed by redefining the phases of the spinons on the two sites. However, the product of $\chi_{ij}$'s around a closed loop

is special. For a plaquette $p$, define

$$e^{i\Phi_p} = \prod_{\langle ij\rangle \in p} \frac{\chi_{ij}}{|\chi_{ij}|}. \tag{2.66}$$

The local phase factors cancel around the loop, so $\Phi_p$ is gauge invariant modulo $2\pi$. This is why a complex pattern of $\chi_{ij}$'s is often called a flux phase: the spinons behave as if they are hopping in a background gauge flux.

This is a useful but slightly dangerous language. At the mean-field level, the spinons appear as free fermions moving in the background set by $\chi_{ij}$, $\Delta_{ij}$, and $\lambda_i$. They may even have a band structure, Dirac points, or a Fermi surface. But this is not an ordinary electronic band structure. The spinons carry spin-$\frac{1}{2}$, but they do not carry electric charge. A spinon Fermi surface, for example, is a Fermi surface of neutral spin excitations, not a metal in the usual sense.

The gauge field enters when we remember that the mean-field parameters should fluctuate. For example, one may write

$$\chi_{ij} = |\chi_{ij}| e^{i a_{ij}}, \tag{2.67}$$

where $a_{ij}$ behaves like a lattice gauge field. Under the local transformation in Eq. (2.62), it transforms as

$$a_{ij} \to a_{ij} + \theta_j - \theta_i. \tag{2.68}$$

Similarly, the Lagrange multiplier $\lambda_i$, which enforces the constraint, plays the role of the temporal component of the gauge field. So the constraint and the gauge field are not separate stories: they are two sides of the same redundancy.

This also explains why mean-field theory must be treated with care. At mean-field level, we freeze $a_{ij}$ and $\lambda_i$ to some saddle-point values. But the real spin system does not contain a fixed background gauge field chosen by hand. The gauge field is dynamical, and its fluctuations can change the physics strongly. If the gauge fluctuations confine the spinons, then the spinons are not physical quasiparticles by themselves; they are forced back into conventional spin-1 excitations such as magnons. If the gauge field is deconfined, then spinons can survive as fractionalized excitations of the spin liquid.

This is the central lesson of the Heisenberg example. The parton construction does not simply turn the Heisenberg model into a free-fermion problem. It gives us a larger language in which several possibilities become visible:

- a hopping ansatz may give dispersing spinons;
- a pairing ansatz may give an RVB-like state of singlet bonds;
- complex hopping amplitudes may generate gauge fluxes;
- gauge fluctuations decide whether the mean-field spinons remain deconfined or become confined.

Thus the mean-field ansatz is only the beginning. The physical question is what survives after projection and gauge fluctuations are taken seriously.

There is one more subtle point. Different mean-field ansätze may look different but actually represent the same physical spin state if they are related by a gauge transformation. Similarly, some phase transitions that appear at the mean-field level may not correspond to real physical phase transitions. They may simply reflect a change in the gauge description. Therefore, when using parton mean-field theory, we must always ask: is this distinction gauge invariant, or is it only a feature of the chosen representation?

For the rest of the report, this warning will stay in the background. In the Heisenberg model, the parton construction teaches us the basic machinery: spinons, constraints, hopping and pairing channels, and gauge redundancy. In the Kitaev model, we will meet a much more controlled situation. There, fractionalization is not merely a mean-field guess; the model can be solved exactly in terms of Majorana fermions and a $\mathbb{Z}_2$ gauge field. Finally, in the Kitaev–Heisenberg model, we return to the less protected situation where the spin-liquid description competes with magnetic order.

## 3 The Kitaev Model: An Exactly Solvable Spin Liquid

### 3.1 The Honeycomb Lattice and Bond-Dependent Exchange

The Kitaev model [2] is a spin-$\frac{1}{2}$ model defined on the honeycomb lattice. At first sight, this looks like a very modest starting point: each site has only three nearest neighbours. The important twist is that the three nearest-neighbour bonds are not treated in the same way. They are labelled as $x$-, $y$-, and $z$-type bonds, and each bond couples only one corresponding component of the spin.

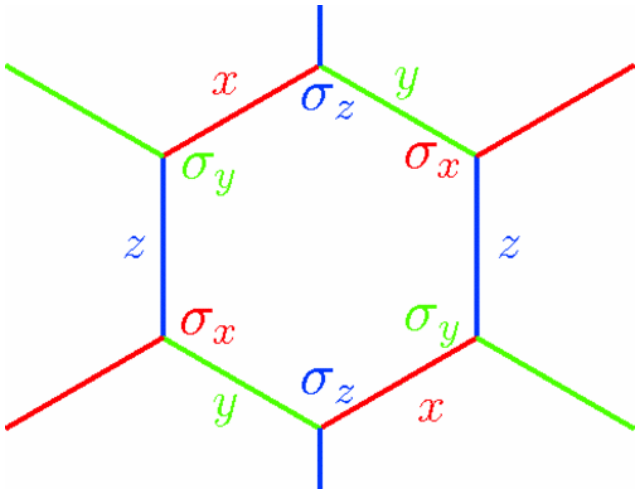



**Figure 4**: Bond-dependent exchange in the Kitaev honeycomb model. The three nearest-neighbour bond types are labelled $x$, $y$, and $z$. On an $x$-bond only the $x$-component of spin is coupled, on a $y$-bond only the $y$-component is coupled, and on a $z$-bond only the $z$-component is coupled. The figure uses Pauli-matrix notation $\sigma^\gamma$, related to spin operators by $S^\gamma = \sigma^\gamma/2$ for spin-$\frac{1}{2}$. Adapted from Burnell and Nayak, Ref. [3].

The Hamiltonian is

$$H_K = -\sum_{\gamma=x,y,z} J_\gamma \sum_{\langle ij\rangle_\gamma} S_i^\gamma S_j^\gamma. \tag{3.1}$$

Here $\langle ij\rangle_\gamma$ denotes a nearest-neighbour bond of type $\gamma$. Thus, on an $x$-bond the interaction is $S_i^x S_j^x$, on a $y$-bond it is $S_i^y S_j^y$, and on a $z$-bond it is $S_i^z S_j^z$. This is very different from the Heisenberg interaction,

$$\mathbf{S}_i \cdot \mathbf{S}_j = S_i^x S_j^x + S_i^y S_j^y + S_i^z S_j^z, \tag{3.2}$$

where every bond treats all spin components equally. In the Kitaev model, the lattice direction decides which spin component is allowed to talk across that bond.

This bond-dependent exchange is the source of the unusual physics. A spin on a honeycomb site has three neighbours. Along one bond its $S^x$ component is involved, along another bond its $S^y$ component is involved, and along the third bond its $S^z$ component is involved. But spin components on the same site do not commute:

$$[S_i^x, S_i^y] = iS_i^z, \qquad [S_i^y, S_i^z] = iS_i^x, \qquad [S_i^z, S_i^x] = iS_i^y. \tag{3.3}$$

So the spin is being asked to satisfy three different, mutually incompatible kinds of local preferences. This is not ordinary geometrical frustration like on a triangular lattice. The honeycomb lattice itself is bipartite. The frustration here comes from the bond-dependent locking of spin space to real space.

The remarkable fact is that this frustrated-looking model is exactly solvable. Kitaev showed that the spin operators can be rewritten in terms of Majorana fermions coupled to a $\mathbb{Z}_2$ gauge field. After this rewriting, the interacting spin problem reduces to free Majorana fermions moving in a static gauge background. This is why the Kitaev model is so important: it gives a rare example where fractionalization is not just a mean-field guess, but an exact result.

The phase structure is also simple enough to state at this stage. When one coupling dominates strongly, for example

$$|J_z| > |J_x| + |J_y|, \tag{3.4}$$

the spectrum is gapped and the system lies in one of the Abelian $A$ phases. When the couplings are more balanced and satisfy the triangle inequalities,

$$|J_x| \leq |J_y| + |J_z|, \qquad |J_y| \leq |J_z| + |J_x|, \qquad |J_z| \leq |J_x| + |J_y|, \tag{3.5}$$

the system lies in the gapless $B$ phase. The isotropic point $J_x = J_y = J_z$ belongs to this gapless region. In both the gapped and gapless regimes, the ground state has no conventional magnetic order. Instead, its excitations fractionalize into itinerant Majorana fermions and $\mathbb{Z}_2$ fluxes.

**A small comment on the description.** The Kitaev model can be discussed using different parton descriptions. In this report, we will follow Kitaev's original Majorana parton route, because it gives the exact solution most directly. There is also a useful SU(2) slave-fermion interpretation in which the Kitaev phases are described as $p$-wave superconducting states of auxiliary fermions. I will not develop that formalism here; for those details, one may refer to Rabi's report [1] and the original work [3].

For this report, the Kitaev model plays the role of a controlled benchmark. The Heisenberg model introduced the general parton logic: enlarge the Hilbert space, introduce auxiliary particles, impose constraints, and worry about gauge redundancy. The Kitaev model now shows a much sharper version of the same story. Here the fractionalized particles and the gauge field appear in an exact solution, so we can see clearly what the parton language is trying to describe. This makes it the natural bridge between the general Heisenberg parton construction and the more realistic Kitaev–Heisenberg model discussed later.

### 3.2 Majorana Partons: Splitting a Spin

The exact solution of the Kitaev model begins with a different kind of parton decomposition. Instead of writing the spin in terms of ordinary complex fermions, Kitaev rewrites it using Majorana fermions. It is convenient to first work with Pauli operators,

$$\sigma_i^\gamma = 2S_i^\gamma, \qquad \gamma = x, y, z. \tag{3.6}$$

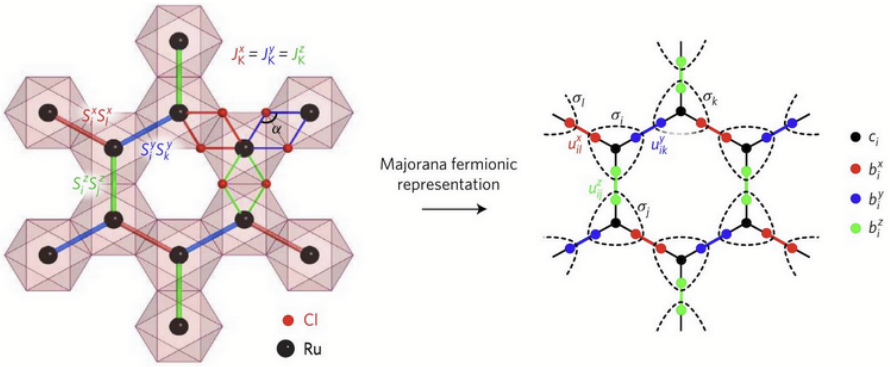



**Figure 5**: Majorana fermionic representation of the Kitaev honeycomb model. The physical spin degrees of freedom on the honeycomb lattice are rewritten in terms of one matter Majorana $c_i$ and three bond Majoranas $b_i^x, b_i^y, b_i^z$. The bond Majoranas combine into $\mathbb{Z}_2$ gauge fields $u_{ij}^\gamma = ib_i^\gamma b_j^\gamma$, while the $c_i$ Majoranas form the itinerant matter sector. This figure visually summarizes the fractionalization $\sigma_i^\gamma = ib_i^\gamma c_i$. Adapted from Ref. [4].

At every lattice site $i$, we introduce four Majorana operators,

$$c_i, \qquad b_i^x, \qquad b_i^y, \qquad b_i^z. \tag{3.7}$$

They are Hermitian,

$$c_i^\dagger = c_i, \qquad (b_i^\gamma)^\dagger = b_i^\gamma, \tag{3.8}$$

and satisfy the Majorana anticommutation relations

$$\{c_i, c_j\} = 2\delta_{ij}, \qquad \{b_i^\gamma, b_j^{\gamma'}\} = 2\delta_{ij}\delta_{\gamma\gamma'}, \qquad \{c_i, b_j^\gamma\} = 0. \tag{3.9}$$

The spin operator is represented as

$$\sigma_i^\gamma = ib_i^\gamma c_i. \tag{3.10}$$

Equivalently, in terms of the spin-$\frac{1}{2}$ operator,

$$S_i^\gamma = \frac{i}{2} b_i^\gamma c_i. \tag{3.11}$$

The notation already hints at the structure of the solution: the $c_i$ Majorana will become the mobile, matter-like degree of freedom, while the three $b_i^\gamma$ Majoranas are tied to the three bond directions.

As before, this rewriting enlarges the Hilbert space. A spin-$\frac{1}{2}$ has only two physical states, while four Majoranas are equivalent to two complex fermionic modes and therefore describe a four-dimensional local Hilbert space. So we have again introduced extra, unphysical states. To remove them, we impose the local constraint

$$D_i = b_i^x b_i^y b_i^z c_i = 1. \tag{3.12}$$

Physical states must satisfy

$$D_i \left|\psi_{\text{phys}}\right\rangle = \left|\psi_{\text{phys}}\right\rangle, \qquad \text{for every site } i. \tag{3.13}$$

This is the Majorana version of the projection back to the physical spin Hilbert space.

Now comes the key simplification. Consider a $\gamma$-type bond $\langle ij\rangle_\gamma$. Using Eq. (3.10),

$$\begin{aligned}\sigma_i^\gamma \sigma_j^\gamma &= (ib_i^\gamma c_i)\left(ib_j^\gamma c_j\right) \\ &= -b_i^\gamma c_i b_j^\gamma c_j.\end{aligned} \tag{3.14}$$

Since $c_i$ anticommutes with $b_j^\gamma$, we can reorder the operators as

$$b_i^\gamma c_i b_j^\gamma c_j = -b_i^\gamma b_j^\gamma c_i c_j. \tag{3.15}$$

Therefore,

$$\sigma_i^\gamma \sigma_j^\gamma = b_i^\gamma b_j^\gamma c_i c_j. \tag{3.16}$$

We now define the bond operator

$$u_{ij} = ib_i^\gamma b_j^\gamma, \qquad \langle ij\rangle_\gamma. \tag{3.17}$$

Since $b_i^\gamma b_j^\gamma = -iu_{ij}$, the spin interaction becomes

$$\sigma_i^\gamma \sigma_j^\gamma = -iu_{ij} c_i c_j. \tag{3.18}$$

If the Kitaev Hamiltonian is written in Pauli-operator convention as

$$H_K = -\sum_{\gamma=x,y,z} K_\gamma \sum_{\langle ij\rangle_\gamma} \sigma_i^\gamma \sigma_j^\gamma, \tag{3.19}$$

then Eq. (3.18) gives

$$H_K = i \sum_{\gamma=x,y,z} K_\gamma \sum_{\langle ij\rangle_\gamma} u_{ij} c_i c_j. \tag{3.20}$$

If instead one writes the Hamiltonian using spin operators $S_i^\gamma = \sigma_i^\gamma/2$, the same expression appears with the coupling rescaled by the corresponding factor of $1/4$.

This is the central miracle of the Kitaev model. The original Hamiltonian is an interacting spin model, but after the Majorana rewriting it becomes a hopping problem for the $c_i$ Majoranas. The hopping signs are controlled by the bond variables $u_{ij}$. These bond variables satisfy

$$u_{ij}^2 = 1, \qquad u_{ij} = \pm 1. \tag{3.21}$$

So the $u_{ij}$'s behave like $\mathbb{Z}_2$ gauge fields living on the bonds of the honeycomb lattice.

In this language, the spin has split into two pieces:

$$\text{spin} \quad \longrightarrow \quad \text{matter Majorana } c_i + \mathbb{Z}_2 \text{ gauge field } u_{ij}. \tag{3.22}$$

This is the exact fractionalization that makes the Kitaev model so important. In the next subsection, we will see why the $u_{ij}$'s are conserved quantities and how their gauge-invariant information is encoded in plaquette fluxes.

### 3.3 $\mathbb{Z}_2$ Gauge Fields and Conserved Fluxes

The Majorana representation turns the Kitaev Hamiltonian into a hopping problem for the matter Majoranas $c_i$. The price is that the hopping amplitudes are not ordinary numbers at first; they contain the bond operators

$$u_{ij} = ib_i^\gamma b_j^\gamma, \qquad \langle ij\rangle_\gamma. \tag{3.23}$$

These variables live on the bonds of the honeycomb lattice. They are Hermitian,

$$u_{ij}^\dagger = u_{ij}, \tag{3.24}$$

and satisfy

$$u_{ij}^2 = 1. \tag{3.25}$$

Therefore their eigenvalues are simply

$$u_{ij} = \pm 1. \tag{3.26}$$

This is the first appearance of the $\mathbb{Z}_2$ gauge structure: each bond carries a two-valued gauge variable.

Using these bond variables, the Kitaev Hamiltonian in the Pauli-operator convention becomes

$$H_K = i \sum_{\gamma=x,y,z} K_\gamma \sum_{\langle ij\rangle_\gamma} u_{ij} c_i c_j. \tag{3.27}$$

The crucial fact is that every $u_{ij}$ commutes with the Hamiltonian,

$$[u_{ij}, H_K] = 0. \tag{3.28}$$

So the $u_{ij}$'s are constants of motion. For any fixed configuration of these bond variables, the Hamiltonian is just a quadratic hopping Hamiltonian for the matter Majoranas $c_i$. This is the key reason the Kitaev model is exactly solvable.

There is, however, one important subtlety. A single $u_{ij}$ is conserved, but it is not itself a physical observable. The Majorana representation has a local gauge redundancy. The local constraint operator

$$D_i = b_i^x b_i^y b_i^z c_i \tag{3.29}$$

generates a local $\mathbb{Z}_2$ gauge transformation. Acting with $D_i$ changes the sign of all four Majoranas at site $i$:

$$c_i \to -c_i, \qquad b_i^\gamma \to -b_i^\gamma. \tag{3.30}$$

As a result, the three bond variables connected to site $i$ change sign, but the physical spin operators and the Hamiltonian remain unchanged. Thus two $u_{ij}$ configurations related by such local sign flips represent the same physical situation.

The physical gauge-invariant information is instead stored in products of $u_{ij}$'s around closed loops. For a hexagonal plaquette $p$, we define the plaquette flux

$$W_p = \prod_{\langle ij\rangle \in \partial p} u_{ij}, \tag{3.31}$$

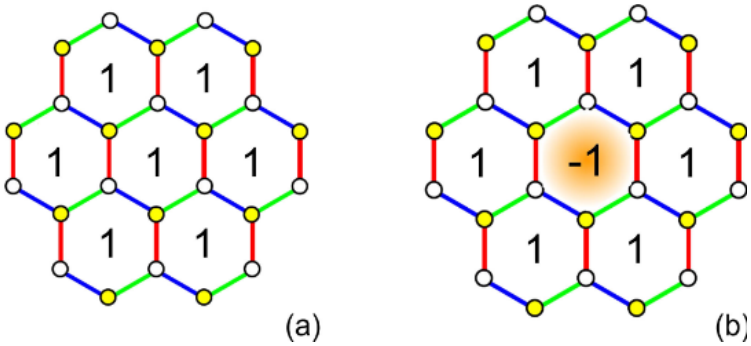



**Figure 6**: Plaquette flux sectors in the Kitaev honeycomb model. The number inside each hexagon denotes the eigenvalue of the gauge-invariant plaquette operator $W_p$. Panel (a) shows the flux-free sector, where $W_p = +1$ on every plaquette. Panel (b) shows a configuration with one plaquette carrying $W_p = -1$, corresponding to a $\mathbb{Z}_2$ flux excitation or vison. Individual bond variables $u_{ij}$ depend on the gauge choice, but the plaquette product $W_p = \prod_{\langle ij \rangle \in \partial p} u_{ij}$ is gauge invariant. Adapted from Ref. [5].

where the product runs around the six bonds forming the boundary of the plaquette, with a fixed orientation convention. Since each $u_{ij} = \pm 1$, the flux also has eigenvalues

$$W_p = \pm 1. \tag{3.32}$$

Unlike a single bond variable, $W_p$ is gauge invariant: the local sign changes cancel when we multiply around a closed loop.

The same flux operator can also be written directly in terms of the original spin variables. For a suitable convention of labelling the six sites around a hexagon,

$$W_p = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z. \tag{3.33}$$

The precise sequence of $x, y, z$ labels depends on the convention used for the plaquette, but the important point is convention-independent: $W_p$ is a product of spin components around a closed loop. It commutes with the Hamiltonian,

$$[W_p, H_K] = 0. \tag{3.34}$$

Therefore the Hilbert space splits into flux sectors labelled by the eigenvalues of all plaquette operators.

For the standard Kitaev model, the ground state lies in the flux-free sector,

$$W_p = +1 \qquad \text{for every plaquette } p. \tag{3.35}$$

A plaquette with

$$W_p = -1 \tag{3.36}$$

contains a $\mathbb{Z}_2$ flux excitation. These fluxes are static in the exactly solvable Kitaev model because all the $W_p$'s commute with $H_K$.

The solution therefore proceeds in two stages. First, choose a flux sector, equivalently a gauge-inequivalent pattern of $u_{ij}$'s. Second, diagonalize the free Majorana Hamiltonian for the $c_i$'s in that background. The physical ground state is obtained by choosing the lowest-energy flux sector and then projecting back to the physical spin Hilbert space.

This separation is the essence of fractionalization in the Kitaev model:

$$\text{spin excitation} \quad \longrightarrow \quad \text{matter Majoranas} + \mathbb{Z}_2 \text{ fluxes.} \tag{3.37}$$

The $c_i$ Majoranas are itinerant, while the $\mathbb{Z}_2$ fluxes label the gauge background through which they move.

### 3.4 Free Majorana Hamiltonian and Spectrum

Once the flux sector is fixed, the Kitaev model becomes a free Majorana hopping problem. For the ground state, we work in the flux-free sector,

$$W_p = +1 \qquad \text{for every plaquette } p. \tag{3.38}$$

A convenient gauge choice is to orient every bond from the $A$-sublattice to the $B$-sublattice and set

$$u_{ij} = +1 \tag{3.39}$$

on these oriented bonds. This does not mean that the gauge field has disappeared; it simply means that we have chosen a representative of the flux-free sector.

The honeycomb lattice has two sites per unit cell, which we label by $A$ and $B$. Let the $z$-bond lie inside the unit cell, while the $x$- and $y$-bonds connect to neighbouring unit cells. With this convention, the Majorana Hamiltonian can be written as

$$H_K = i \sum_{\mathbf{R}} \left[ K_z c_{A,\mathbf{R}} c_{B,\mathbf{R}} + K_x c_{A,\mathbf{R}} c_{B,\mathbf{R}+\mathbf{a}_1} + K_y c_{A,\mathbf{R}} c_{B,\mathbf{R}+\mathbf{a}_2} \right], \tag{3.40}$$

where $\mathbf{R}$ labels the unit cell and $\mathbf{a}_1, \mathbf{a}_2$ are Bravais lattice vectors. The precise definition of $\mathbf{a}_1, \mathbf{a}_2$ depends on the chosen unit cell, but the physics does not.

We now Fourier transform the Majorana operators:

$$c_{s,\mathbf{R}} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{R}} c_{s,\mathbf{k}}, \qquad s = A, B. \tag{3.41}$$

Since $c_{s,\mathbf{R}}$ is a Majorana operator, it satisfies

$$c^\dagger_{s,\mathbf{k}} = c_{s,-\mathbf{k}}. \tag{3.42}$$

Substituting the Fourier expansion into Eq. (3.40), the Hamiltonian becomes

$$H_K = \frac{i}{2} \sum_{\mathbf{k}} \begin{pmatrix} c_{A,-\mathbf{k}} & c_{B,-\mathbf{k}} \end{pmatrix} \begin{pmatrix} 0 & \Phi(\mathbf{k}) \\ -\Phi^*(\mathbf{k}) & 0 \end{pmatrix} \begin{pmatrix} c_{A,\mathbf{k}} \\ c_{B,\mathbf{k}} \end{pmatrix}, \tag{3.43}$$

where

$$\Phi(\mathbf{k}) = K_z + K_x e^{i\mathbf{k}\cdot\mathbf{a}_1} + K_y e^{i\mathbf{k}\cdot\mathbf{a}_2}. \tag{3.44}$$

Thus the many-body spin problem has reduced to diagonalizing a $2 \times 2$ matrix at each momentum. The positive Majorana excitation energy is

$$E(\mathbf{k}) = 2|\Phi(\mathbf{k})|, \tag{3.45}$$

up to the overall normalization chosen for the Kitaev couplings. The exact prefactor is convention-dependent; the important physics is contained in whether $\Phi(\mathbf{k})$ can vanish.

The system is gapless when

$$\Phi(\mathbf{k}) = 0 \tag{3.46}$$

has a solution inside the Brillouin zone. Geometrically, this is possible only when the three coupling magnitudes can form the sides of a triangle:

$$|K_x| \leq |K_y| + |K_z|, \qquad |K_y| \leq |K_z| + |K_x|, \qquad |K_z| \leq |K_x| + |K_y|. \tag{3.47}$$

When these inequalities are satisfied, the model lies in the gapless $B$ phase.

At the isotropic point,

$$K_x = K_y = K_z \equiv K, \tag{3.48}$$

we have

$$\Phi(\mathbf{k}) = K\left(1 + e^{i\mathbf{k}\cdot\mathbf{a}_1} + e^{i\mathbf{k}\cdot\mathbf{a}_2}\right). \tag{3.49}$$

This function vanishes at two inequivalent momenta, usually denoted by $\mathbf{K}$ and $\mathbf{K}'$, analogous to the Dirac points of graphene. Expanding near one of these points,

$$\mathbf{k} = \mathbf{K} + \mathbf{q}, \qquad |\mathbf{q}| \ll 1, \tag{3.50}$$

one obtains a linear spectrum,

$$E(\mathbf{K} + \mathbf{q}) \sim v|\mathbf{q}|, \tag{3.51}$$

where $v$ is the Majorana velocity. Thus the low-energy excitations in the isotropic Kitaev model are gapless Majorana fermions with Dirac-cone-like dispersion.

If one coupling dominates, for example

$$|K_z| > |K_x| + |K_y|, \tag{3.52}$$

then $\Phi(\mathbf{k})$ cannot vanish anywhere in the Brillouin zone. The spectrum is fully gapped. This is one of the Abelian $A$ phases of the Kitaev model.

The spectrum gives a direct way to understand the phase structure. The $B$ phase contains gapless itinerant Majorana fermions moving in the flux-free $\mathbb{Z}_2$ gauge background. The $A$ phase is gapped and is smoothly connected to a toric-code-like Abelian topological phase. In both phases, the original spin degrees of freedom have fractionalized: the mobile part is carried by the matter Majoranas, while the $\mathbb{Z}_2$ fluxes describe the gauge sector.

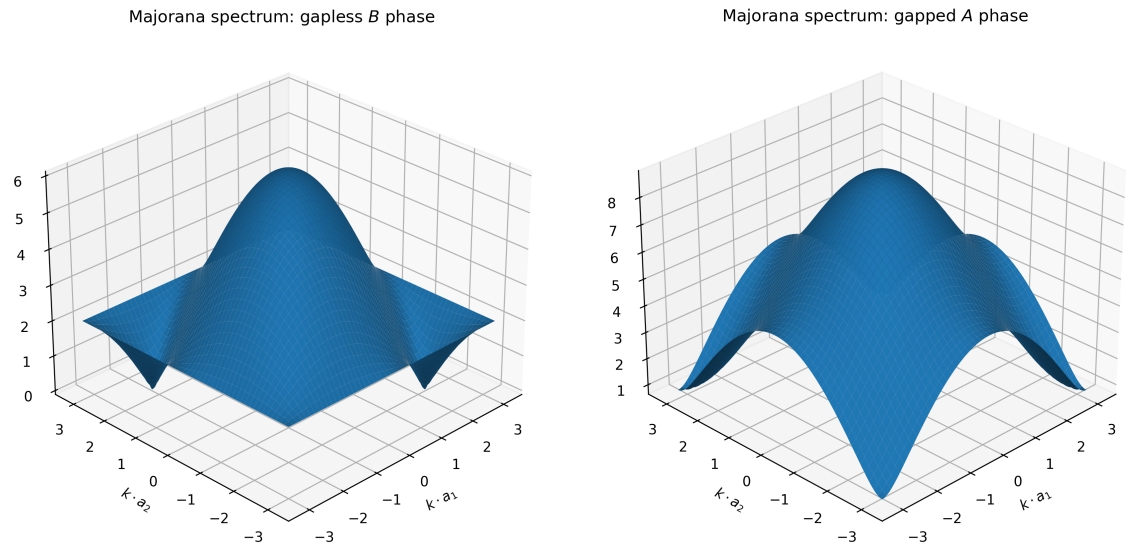



**Figure 7**: Positive Majorana excitation spectrum of the Kitaev model in the flux-free sector. Left: the isotropic point $K_x = K_y = K_z$, where $E(\mathbf{k}) = 2|\Phi(\mathbf{k})|$ vanishes at two Dirac-cone-like nodes, corresponding to the gapless $B$ phase. Right: an anisotropic point with $|K_z| > |K_x| + |K_y|$, where the positive excitation spectrum is fully gapped, corresponding to an Abelian $A$ phase. The full Majorana/BdG spectrum contains the symmetric branches $\pm E(\mathbf{k})$; here only the positive excitation branch is shown. The axes use reduced momentum coordinates $\mathbf{k} \cdot \mathbf{a}_1$ and $\mathbf{k} \cdot \mathbf{a}_2$. Plots generated numerically.

### 3.5 What Makes the Kitaev Model a Spin Liquid?

The Kitaev model is important because it gives an exactly solvable example of a quantum spin liquid. The ground state has no conventional magnetic order, but it is not a trivial disordered state. Instead, the spin degrees of freedom split into two different kinds of excitations:

$$\text{spin} \quad \longrightarrow \quad \text{matter Majoranas} + \mathbb{Z}_2 \text{ fluxes.} \tag{3.53}$$

The matter Majoranas $c_i$ are itinerant particles which form the band structure discussed above, while the $\mathbb{Z}_2$ fluxes are gauge excitations living on the plaquettes. This is the central reason why the Kitaev model is not just another disordered magnet: its elementary excitations are not the original spins.

This fractionalization also explains why the usual magnon picture is not suitable here. In an ordered magnet, a spin flip creates a magnon, which is a spin-one excitation moving through an ordered background. In the Kitaev spin liquid, there is no ordered background. A local spin operator changes both the matter Majorana sector and the flux sector. Equivalently, a spin flip injects a Majorana excitation and also changes the nearby $\mathbb{Z}_2$ gauge flux configuration. So a spin excitation does not behave like a single sharp quasiparticle. Instead of a clean magnon branch, one expects a broad continuum in dynamical probes such as neutron scattering.

A special feature of the exactly solvable point is that the $\mathbb{Z}_2$ fluxes are static. Since every plaquette operator $W_p$ commutes with the Hamiltonian,

$$[W_p, H_K] = 0, \tag{3.54}$$

the flux configuration is conserved. The ground state is flux-free,

$$W_p = +1 \qquad \text{for every plaquette } p, \tag{3.55}$$

while a plaquette with

$$W_p = -1 \tag{3.56}$$

contains a $\mathbb{Z}_2$ flux excitation, often called a vison. Thus the spin liquid has two very different kinds of degrees of freedom: fast itinerant matter Majoranas and slower, gapped flux excitations.

The physical spin correlations are also unusual. For the pure Kitaev model, equal-time spin correlations are extremely short-ranged. The reason is simple but important: acting with a single spin operator changes the flux configuration on nearby plaquettes. For a nonzero spin correlation, the fluxes created by one spin operator must be removed by the other. This severely restricts where the second spin operator can act. As a result, the model has no magnetic long-range order, even though the system is very far from being a collection of independent spins.

The two main phases of the Kitaev model are the gapped $A$ phase and the gapless $B$ phase. In the $A$ phase, one coupling dominates, for example

$$|K_z| > |K_x| + |K_y|. \tag{3.57}$$

The Majorana spectrum is fully gapped, and this phase is smoothly connected to a toric-code-like Abelian $\mathbb{Z}_2$ topological order. In the $B$ phase, the couplings satisfy the triangle inequalities,

$$|K_x| \le |K_y| + |K_z|, \qquad |K_y| \le |K_z| + |K_x|, \qquad |K_z| \le |K_x| + |K_y|. \tag{3.58}$$

The matter Majoranas are gapless and have Dirac-cone-like nodes. The isotropic point

$$K_x = K_y = K_z \tag{3.59}$$

lies inside this gapless phase.

A further important result is that the gapless $B$ phase can become a gapped non-Abelian spin liquid when time-reversal symmetry is broken, for example by an effective magnetic-field-induced three-spin term. This perturbation opens a gap in the Majorana spectrum and produces a phase supporting non-Abelian Ising anyons. This is one of the reasons the Kitaev model became so famous: it is not only exactly solvable, but also sits very close to ideas from topological quantum computation.

The thermodynamic properties also reflect fractionalization. Because the spin effectively splits into matter Majoranas and $\mathbb{Z}_2$ fluxes, the entropy is released in two stages

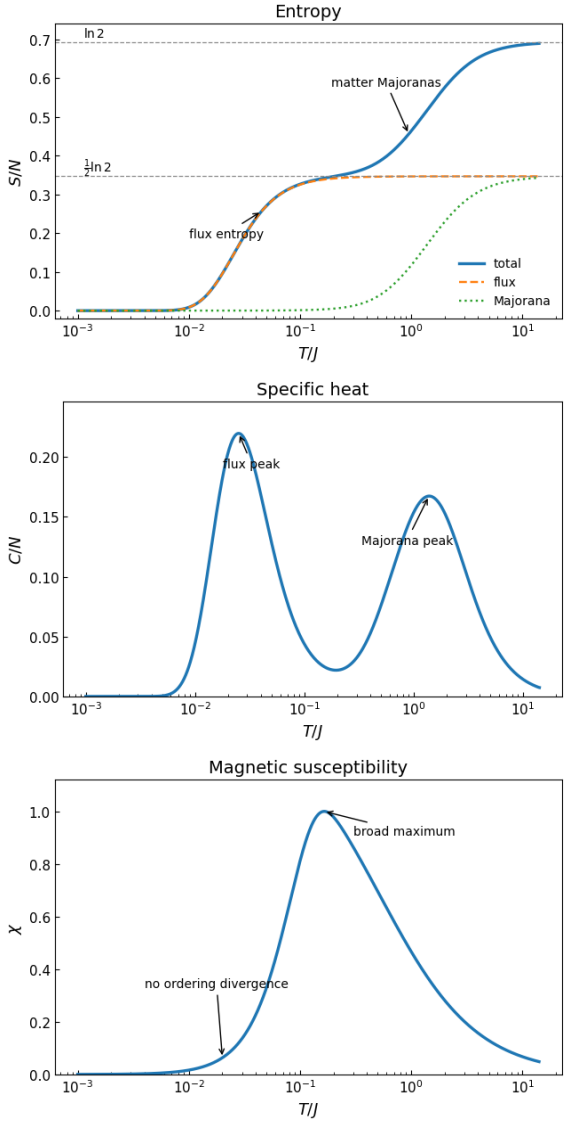



**Figure 8**: Thermodynamic signatures of the Kitaev spin liquid. Top: entropy per site $S/N$, showing a two-stage release associated with the $\mathbb{Z}_2$ flux sector and the itinerant matter-Majorana sector. Middle: specific heat per site $C/N$, displaying two broad peaks corresponding to these two energy scales. Bottom: magnetic susceptibility $\chi$, showing a broad maximum without a divergence associated with conventional magnetic ordering. (These curves are meant as illustrative numerical diagnostics rather than an exact finite-temperature solution.)

rather than all at once. At a higher temperature scale, the itinerant Majorana sector becomes correlated. At a lower temperature scale, the flux sector freezes into the flux-free configuration. Correspondingly, the specific heat shows two broad features rather than a sharp thermal phase transition. This is a useful diagnostic: the Kitaev spin liquid does not order thermally in the conventional sense, but it still has clear crossover signatures

associated with its fractionalized degrees of freedom.

The magnetic response has a similar lesson. Since there is no conventional ordered moment, the susceptibility does not signal a transition into a magnetically ordered phase. Instead, it reflects the buildup of short-range spin correlations and the fractionalized excitation structure. Likewise, the dynamical spin structure factor is broad and continuum-like, rather than being dominated by sharp magnon modes. This continuum is often taken as one of the key experimental fingerprints of fractionalization in Kitaev-like materials, although in real materials it must be interpreted carefully because additional interactions can broaden, shift, or partially confine the ideal Kitaev excitations.

Thus the Kitaev model provides the clean benchmark for the rest of the report. It shows explicitly how a spin system can avoid magnetic order, fractionalize into emergent particles, and produce a $\mathbb{Z}_2$ gauge structure. The Kitaev–Heisenberg model can then be viewed as the next step: what happens when this exactly solvable spin liquid is perturbed by ordinary Heisenberg exchange?

# 4 The Kitaev–Heisenberg Model

## 4.1 The Effective Spin Hamiltonian

The Kitaev–Heisenberg model is the natural next step after the pure Kitaev model. The Kitaev model by itself is exactly solvable and gives a clean spin-liquid ground state, but it is also a very special limit. In a more realistic spin–orbit-coupled magnetic system, the bond-dependent Kitaev interaction will generally appear together with more ordinary exchange terms. The simplest way to study this competition is to keep both the Kitaev and Heisenberg interactions on the honeycomb lattice.

The model studied here is the Kitaev–Heisenberg model introduced in the paper by Jirí Chaloupka, George Jackeli, and Giniyat Khaliullin [6],

$$H = (1-\alpha)H_H - 2\alpha H_K, \qquad 0 \le \alpha \le 1, \tag{4.1}$$

where

$$H_H = \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j \tag{4.2}$$

is the nearest-neighbour antiferromagnetic Heisenberg interaction, and

$$H_K = \sum_{\gamma=x,y,z} \sum_{\langle ij \rangle_\gamma} S_i^\gamma S_j^\gamma \tag{4.3}$$

is the bond-dependent Kitaev interaction. Here $\langle ij \rangle_\gamma$ denotes a nearest-neighbour bond of type $\gamma = x, y, z$. Notice that in this section $H_K$ denotes only the positive bond operator $\sum_{\langle ij \rangle_\gamma} S_i^\gamma S_j^\gamma$; the ferromagnetic sign is included explicitly through the coefficient $-2\alpha$ in Eq. (4.1).

Thus, on an $x$-bond the Kitaev term couples only $S_i^x S_j^x$, on a $y$-bond only $S_i^y S_j^y$, and on a $z$-bond only $S_i^z S_j^z$.

The parameter $\alpha$ controls which tendency dominates. At

$$\alpha = 0, \tag{4.4}$$

the Hamiltonian becomes

$$H(\alpha = 0) = H_H. \tag{4.5}$$

This is the antiferromagnetic Heisenberg model on the honeycomb lattice. Since the honeycomb lattice is bipartite, this limit favours the usual two-sublattice Néel antiferromagnetic order.

At the other end,

$$\alpha = 1, \tag{4.6}$$

the Hamiltonian becomes

$$H(\alpha = 1) = -2H_K. \tag{4.7}$$

This is the ferromagnetic Kitaev limit. It is continuously connected to the exactly solvable Kitaev spin liquid discussed in the previous section, where the spin degrees of freedom fractionalize into itinerant Majorana fermions and $\mathbb{Z}_2$ fluxes.

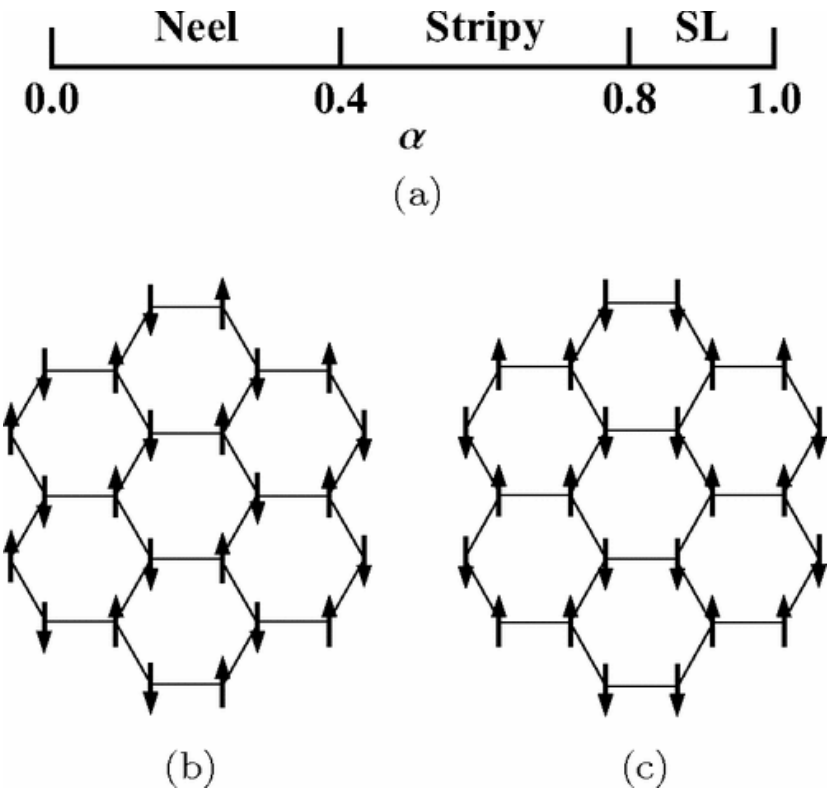



**Figure 9**: Schematic phase diagram of the Kitaev–Heisenberg model and the corresponding magnetic patterns. Panel (a) shows the evolution with the tuning parameter $\alpha$: the system starts from the Néel antiferromagnet near the Heisenberg limit, enters an intermediate stripy antiferromagnetic phase, and finally reaches the spin-liquid regime near the Kitaev limit. Panels (b) and (c) show the Néel and stripy spin configurations on the honeycomb lattice. Adapted from Ref. [7].

So the model puts two very different magnetic instincts into one simple Hamiltonian. The Heisenberg part wants the spins to choose a conventional ordered pattern. The Kitaev part ties spin components to bond directions and favours the fractionalized spin-liquid physics of the Kitaev limit. The interesting question is what happens between these two endpoints.

One might first expect a direct evolution from Néel order to the Kitaev spin liquid. The actual phase diagram is richer. As $\alpha$ is increased, the system first leaves the Néel antiferromagnet and enters an intermediate stripy antiferromagnetic phase before reaching the spin-liquid regime near the Kitaev limit. This stripy phase will become much more transparent after a four-sublattice rotation, which is one of the main tricks used in this model.

In this report, Eq. (4.1) is the central spin Hamiltonian. The parton mean-field viewpoint will help us understand why the Kitaev side naturally supports fractionalized spin-liquid physics, while the four-sublattice rotation will explain why the stripy phase appears so naturally in the middle of the phase diagram. After that, we use numerical diagonaliza-

tion of this Hamiltonian to reproduce the main phase-diagram signatures.

### 4.2 Parton Mean-Field Viewpoint on the Kitaev–Heisenberg Model

The Kitaev–Heisenberg Hamiltonian is difficult because the two ends of the phase diagram prefer very different descriptions. Near the Kitaev limit, the natural language is fractionalized: spins split into emergent fermionic degrees of freedom and gauge variables. Near the ordered side, however, the natural language is magnetic order. A useful parton mean-field theory should therefore be able to keep both possibilities visible at the same time.

Following the slave-particle treatment of the Heisenberg–Kitaev model, it is convenient to work in the rotated basis and introduce

$$\delta = \alpha - \frac{1}{2}. \tag{4.8}$$

The four-sublattice rotation will be explained geometrically in the next subsection. For now, the important result is that the Hamiltonian becomes

$$H = -\left(\frac{1}{2} - \delta\right) H_H - 4\delta H_K, \qquad 0 \le \delta \le \frac{1}{2}. \tag{4.9}$$

Here $\delta = 0$ corresponds to the hidden ferromagnetic point, which is the stripy antiferromagnet in the original basis, while $\delta = \frac{1}{2}$ corresponds to the Kitaev spin-liquid limit. This rotated form is especially useful because both terms are ferromagnetic in the region between the stripy phase and the Kitaev limit.

We now write the spin operator in terms of fermionic spinons,

$$S_i^\mu = \frac{1}{2} f_{i\alpha}^\dagger \sigma_{\alpha\beta}^\mu f_{i\beta}, \qquad \mu = x, y, z, \tag{4.10}$$

with the physical constraint

$$f_{i\uparrow}^\dagger f_{i\uparrow} + f_{i\downarrow}^\dagger f_{i\downarrow} = 1. \tag{4.11}$$

This turns every spin-spin interaction into a four-fermion interaction. The purpose of mean-field theory is to replace these quartic terms by quadratic terms in suitable hopping, pairing, and magnetic channels.

Because the Kitaev interaction is spin anisotropic, singlet channels alone are not enough. The useful decoupling channels are spin-triplet particle-hole and particle-particle fields. For a bond from site $i$ to $i + p$, where $p = x, y, z$ labels the bond type, define [7]

$$E_{i,p}^\mu = \frac{1}{2} f_{i+p,\alpha}^\dagger \sigma_{\alpha\beta}^\mu f_{i\beta}, \tag{4.12}$$

and

$$D_{i,p}^\mu = \frac{1}{2} f_{i+p,\alpha} \left(i\sigma^y \sigma^\mu\right)_{\alpha\beta} f_{i\beta}. \tag{4.13}$$

The field $E_{i,p}^\mu$ is a triplet hopping channel, while $D_{i,p}^\mu$ is a triplet pairing channel. These are the natural mean-field variables for a spin-orbit-coupled model, because they allow the spin direction $\mu$ and the bond direction $p$ to remain explicitly tied to one another.

A useful identity for the component interaction is

$$S_i^\alpha S_{i+p}^\alpha = \frac{1}{2}\sum_{\beta=x,y,z}(1-\delta_{\alpha\beta})\left[\left(E_{i,p}^\beta\right)^\dagger E_{i,p}^\beta + \left(D_{i,p}^\beta\right)^\dagger D_{i,p}^\beta\right] - \frac{n_i}{4}, \tag{4.14}$$

where

$$n_i = \sum_\alpha f_{i\alpha}^\dagger f_{i\alpha}. \tag{4.15}$$

This equation shows why the Kitaev term naturally produces triplet mean fields. On a bond of type $p$, the Kitaev interaction selects one spin component, but the corresponding parton decoupling is expressed through the transverse triplet hopping and pairing channels.

The Heisenberg term contains all three spin components. Using the same decoupling, its mean-field contribution may be written as

$$\begin{aligned} H_H^{\mathrm{MF}} = \frac{1}{4}\sum_{i,p}\Big[ & m\left(f_i^\dagger\sigma^z f_i + f_{i+p}^\dagger\sigma^z f_{i+p}\right) - 2m^2 \\ & + \left(f_i^\dagger \mathbf{E}_{i,p}\cdot\boldsymbol{\sigma} f_{i+p} + \mathrm{H.c.}\right) - 2|\mathbf{E}_{i,p}|^2 \\ & + \left(f_i^\dagger \mathbf{D}_{i,p}\cdot(-i\boldsymbol{\sigma}\sigma^y) f_{i+p}^\dagger + \mathrm{H.c.}\right) - 2|\mathbf{D}_{i,p}|^2\Big]. \end{aligned} \tag{4.16}$$

Here $m$ is the magnetic mean field,

$$m = \frac{1}{2}\left\langle f_i^\dagger\sigma^z f_i\right\rangle, \tag{4.17}$$

which is included so that the theory can also describe the magnetically ordered side. Without this channel, the mean-field theory would be biased toward spin liquids only.

For the Kitaev term, the bond type matters explicitly. On a bond of type $p$, the Kitaev interaction couples $S_i^p S_{i+p}^p$. Using Eq. (4.14), the corresponding mean-field Hamiltonian is

$$\begin{aligned} H_K^{\mathrm{MF}} = \frac{1}{4}\sum_{i,p,r}(1-\delta_{pr})\Big[ & \left(f_i^\dagger E_{i,p}^r\sigma^r f_{i+p} + \mathrm{H.c.}\right) - 2|E_{i,p}^r|^2 \\ & + \left(f_i^\dagger D_{i,p}^r(-i\sigma^r\sigma^y) f_{i+p}^\dagger + \mathrm{H.c.}\right) - 2|D_{i,p}^r|^2\Big]. \end{aligned} \tag{4.18}$$

The factor $(1-\delta_{pr})$ is important. It says that on a $p$-type bond, the decoupling involves the triplet channels transverse to $p$. This is the parton mean-field version of the bond-dependent anisotropy of the Kitaev interaction.

Combining the two pieces, the mean-field Hamiltonian corresponding to Eq. (4.9) is

$$H_{\mathrm{MF}} = -\left(\frac{1}{2}-\delta\right)H_H^{\mathrm{MF}} - 4\delta H_K^{\mathrm{MF}} + H_\lambda, \tag{4.19}$$

where

$$H_\lambda = \sum_i \lambda_i\left(f_{i\uparrow}^\dagger f_{i\uparrow} + f_{i\downarrow}^\dagger f_{i\downarrow} - 1\right) \tag{4.20}$$

enforces the single-occupancy constraint at the mean-field level.

For a translationally invariant ansatz, the fields $E^r_{i,p}$, $D^r_{i,p}$, $m$, and $\lambda_i$ are chosen independent of the unit cell, with only their bond and spin labels retained. The Hamiltonian can then be Fourier transformed and written as a Bogoliubov–de Gennes Hamiltonian for spinons:

$$H_{\mathrm{MF}} = C + \sum_{\mathbf{k}} \Psi^\dagger_{\mathbf{k}} \mathcal{H}_{\mathrm{BdG}}(\mathbf{k}) \Psi_{\mathbf{k}}, \tag{4.21}$$

where $C$ contains the constant mean-field energy terms, and the Nambu spinor contains particle and hole components on the two honeycomb sublattices. A convenient choice is

$$\Psi^\dagger_{\mathbf{k},\sigma} = \left( f^\dagger_{\mathbf{k},A,\sigma}, f^\dagger_{\mathbf{k},B,\sigma}, f_{-\mathbf{k},A,\sigma}, f_{-\mathbf{k},B,\sigma} \right). \tag{4.22}$$

The self-consistent mean-field solution is obtained by diagonalizing $\mathcal{H}_{\mathrm{BdG}}(\mathbf{k})$, filling the negative-energy spinon states, and recomputing the expectation values of $E$, $D$, $m$, and $\lambda$ until convergence.

At the Kitaev limit, this construction reproduces the expected fractionalized structure. The complex spinons can be reorganized into four Majorana modes. One Majorana sector is dispersive and gives the gapless matter Majoranas, while the remaining sectors are connected to the $\mathbb{Z}_2$ gauge structure. Thus the parton mean-field picture agrees with the exact Kitaev solution at the endpoint.

Away from the Kitaev limit, the Heisenberg term competes with this spin-liquid structure. In the mean-field description, this competition appears as a competition between the spin-liquid amplitudes $E, D$ and the magnetic order parameter $m$. When $m = 0$, the mean-field ansatz describes a $Z_2$ spin liquid connected to the Kitaev phase. When $m \neq 0$, magnetic order appears. After undoing the four-sublattice rotation, this magnetic order is interpreted as the stripy antiferromagnet of the original Kitaev–Heisenberg model.

For our report, this mean-field construction gives the theoretical bridge between the exactly solvable Kitaev spin liquid and the ordered stripy phase. The numerical reproduction later will work directly with the spin Hamiltonian, but this parton viewpoint explains what is being competed: deconfined spin-liquid amplitudes on one side and magnetic order on the other.

### 4.3 Four-Sublattice Rotation and the Stripy Phase

The intermediate stripy phase looks a little surprising if we stare only at the original Hamiltonian. The Heisenberg limit wants Néel order, the Kitaev limit wants a spin liquid, and yet the model develops a third phase in between. The reason this phase is not accidental becomes clear after a special four-sublattice rotation.

The idea is to divide the honeycomb lattice into four sublattices. On one sublattice, the spins are left unchanged. On the other three sublattices, the spins are rotated by $\pi$ about one of the spin axes:

$$R_x(\pi): \quad (S^x, S^y, S^z) \longrightarrow (S^x, -S^y, -S^z), \tag{4.23}$$

$$R_y(\pi): \quad (S^x, S^y, S^z) \longrightarrow (-S^x, S^y, -S^z), \tag{4.24}$$

$$R_z(\pi): \quad (S^x, S^y, S^z) \longrightarrow (-S^x, -S^y, S^z). \tag{4.25}$$

We denote the spins in this rotated basis by $\widetilde{\mathbf{S}}_i$. This is not a physical approximation; it is just a change of spin variables. The clever part is the choice of rotation pattern. It is chosen so that, on a bond of type $\gamma$, the $\gamma$-component of spin keeps the same relative sign, while the other two components pick up a minus sign.

Consider a $\gamma$-type bond $\langle ij\rangle_\gamma$. Because the $\gamma$-component is unchanged in relative sign on that bond,

$$S_i^\gamma S_j^\gamma = \widetilde{S}_i^\gamma \widetilde{S}_j^\gamma. \tag{4.26}$$

So the Kitaev term is invariant under the four-sublattice rotation:

$$H_K \longrightarrow H_K. \tag{4.27}$$

The Heisenberg term changes more interestingly. On the same $\gamma$-type bond, the two spin components perpendicular to $\gamma$ change sign relative to each other, while the $\gamma$-component does not. Therefore,

$$\begin{aligned} \mathbf{S}_i \cdot \mathbf{S}_j &= S_i^x S_j^x + S_i^y S_j^y + S_i^z S_j^z \\ &\longrightarrow \widetilde{S}_i^\gamma \widetilde{S}_j^\gamma - \sum_{\mu\neq\gamma} \widetilde{S}_i^\mu \widetilde{S}_j^\mu. \end{aligned} \tag{4.28}$$

This can be rewritten as

$$\mathbf{S}_i \cdot \mathbf{S}_j \longrightarrow -\widetilde{\mathbf{S}}_i \cdot \widetilde{\mathbf{S}}_j + 2\widetilde{S}_i^\gamma \widetilde{S}_j^\gamma. \tag{4.29}$$

In Hamiltonian language, this means

$$H_H \longrightarrow -H_H + 2H_K, \qquad H_K \longrightarrow H_K, \tag{4.30}$$

where, for simplicity, the symbols on the right-hand side now refer to the rotated spin variables.

Substituting this into the Kitaev–Heisenberg Hamiltonian,

$$H = (1-\alpha)H_H - 2\alpha H_K, \tag{4.31}$$

we get

$$\begin{aligned} H &\longrightarrow (1-\alpha)(-H_H + 2H_K) - 2\alpha H_K \\ &= -(1-\alpha)H_H + 2(1-2\alpha)H_K. \end{aligned} \tag{4.32}$$

Equivalently,

$$H = -(1-\alpha)H_H - 4\left(\alpha - \frac{1}{2}\right) H_K. \tag{4.33}$$

If we define

$$\delta = \alpha - \frac{1}{2}, \tag{4.34}$$

then this becomes

$$H = -\left(\frac{1}{2} - \delta\right) H_H - 4\delta H_K. \tag{4.35}$$

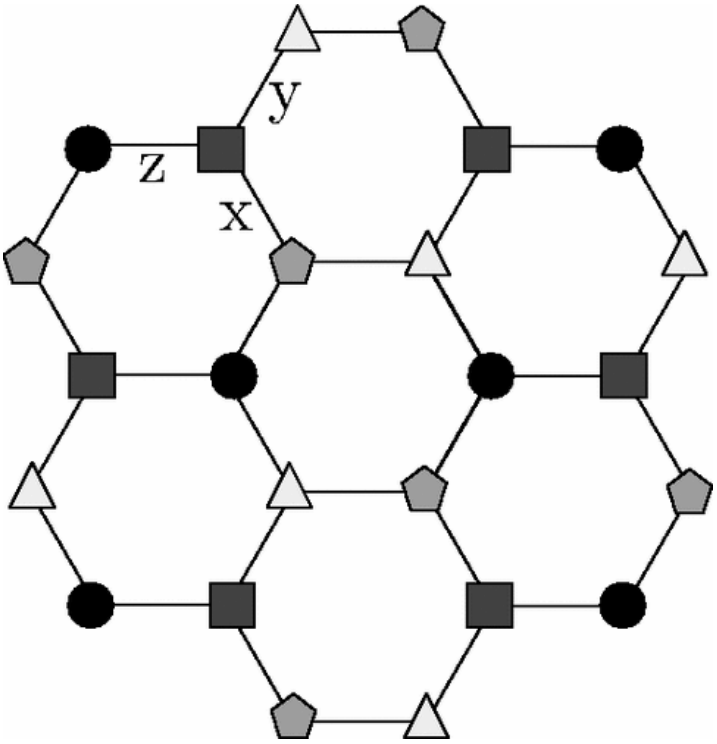



**Figure 10**: Four-sublattice rotation used in the Kitaev–Heisenberg model. The different symbols mark the four sublattices: one sublattice is left invariant, while the other three are rotated by $\pi$ about the $x$, $y$, or $z$ spin axes. Under this transformation, the Kitaev term remains invariant, while the Heisenberg term transforms as $H_H \to -H_H + 2H_K$. Therefore, at $\alpha = 1/2$, the rotated Hamiltonian becomes a simple ferromagnetic Heisenberg model. Transforming back to the original basis gives the stripy antiferromagnetic state. Adapted from Ref. [7].

This is the rotated-basis form used to study the region between the stripy phase and the Kitaev spin liquid.

The special point is now obvious. At

$$\alpha = \frac{1}{2} \qquad \text{or equivalently} \qquad \delta = 0, \tag{4.36}$$

the Kitaev term in Eq. (4.33) disappears. The Hamiltonian reduces to

$$H_{\alpha=1/2} = -\frac{1}{2} H_H. \tag{4.37}$$

This is just a ferromagnetic Heisenberg model in the rotated basis. Its ground state is therefore a simple ferromagnet:

$$\widetilde{\mathbf{S}}_i = \widetilde{\mathbf{S}}_j \qquad \text{for all nearest neighbours.} \tag{4.38}$$

When we transform back to the original spin basis, this rotated ferromagnet becomes the stripy antiferromagnet. So the stripy phase is not a random extra phase inserted

between Néel order and the spin liquid. It is a hidden ferromagnet revealed by a sublattice-dependent spin rotation.

This rotation is also very useful for the numerical work. In the original spin basis, detecting stripy order requires looking for a particular antiferromagnetic pattern. In the rotated basis, the same order appears as ferromagnetism. So one can diagnose the stripy phase by measuring ferromagnetic correlations or the total spin in the rotated variables. This is why the four-sublattice rotation is central both to the analytic understanding of the $\alpha = \frac{1}{2}$ point and to the numerical identification of the intermediate phase.

### 4.4 Numerical Reproduction of the Phase Diagram

The details of the numerical implementation can be found in Uday's report [1]. Here I only describe the basic idea and the physical meaning of the plots. The calculation directly diagonalizes the finite-cluster Kitaev–Heisenberg Hamiltonian

$$H(\alpha) = \sum_{\langle ij\rangle_\gamma} \left[ (1-\alpha)\, \mathbf{S}_i \cdot \mathbf{S}_j - 2\alpha\, S_i^\gamma S_j^\gamma \right], \qquad 0 \le \alpha \le 1. \tag{4.39}$$

For each value of $\alpha$, the ground state is obtained as

$$H(\alpha)\,|\psi_0(\alpha)\rangle = E_{\mathrm{GS}}(\alpha)\,|\psi_0(\alpha)\rangle\,, \tag{4.40}$$

and the relevant observables are evaluated in this ground state. Since the calculation is done on a finite cluster (finite periodic honeycomb cluster of $N = 16$ spin-$\frac{1}{2}$ sites), the results should be read as finite-size signatures of the phases rather than sharp thermodynamic phase transitions.

The first diagnostic is the normalized total-spin square. In the original basis one computes

$$\left\langle \mathbf{S}_{\mathrm{tot}}^2 \right\rangle, \qquad \mathbf{S}_{\mathrm{tot}} = \sum_i \mathbf{S}_i, \tag{4.41}$$

while in the four-sublattice rotated basis one computes

$$\left\langle \widetilde{\mathbf{S}}_{\mathrm{tot}}^2 \right\rangle, \qquad \widetilde{\mathbf{S}}_{\mathrm{tot}} = \sum_i \widetilde{\mathbf{S}}_i. \tag{4.42}$$

This is especially useful because the stripy antiferromagnet becomes a ferromagnet after the four-sublattice rotation. Therefore, a large value of $\langle \widetilde{\mathbf{S}}_{\mathrm{tot}}^2 \rangle$ signals the stripy region. This is exactly what is seen in Fig. 11(a): the rotated total spin grows strongly after the Néel region and becomes largest inside the stripy phase.

The nearest-neighbour correlations give a second view of the same physics. The plotted quantities are the original-basis correlation $C_{\mathrm{NN}}$, the rotated-basis correlation $\widetilde{C}_{\mathrm{NN}}$, and the bond-selective Kitaev correlation $C_K$:

$$C_{\mathrm{NN}} = \frac{1}{N_b} \sum_{\langle ij\rangle} \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle\,, \qquad \widetilde{C}_{\mathrm{NN}} = \frac{1}{N_b} \sum_{\langle ij\rangle} \left\langle \widetilde{\mathbf{S}}_i \cdot \widetilde{\mathbf{S}}_j \right\rangle, \tag{4.43}$$

and

$$C_K = \frac{1}{N_b} \sum_{\langle ij\rangle_\gamma} \left\langle S_i^\gamma S_j^\gamma \right\rangle. \tag{4.44}$$

In Fig. 11(b), $C_{\mathrm{NN}}$ is strongly negative near the Heisenberg limit, consistent with antiferromagnetic correlations. After the transition into the stripy region, $\widetilde{C}_{\mathrm{NN}}$ becomes large and positive, again showing that the stripy phase is ferromagnetic in the rotated basis. Closer to the Kitaev limit, the ordinary magnetic correlations weaken, while the bond-selective Kitaev correlation remains the more natural local quantity.

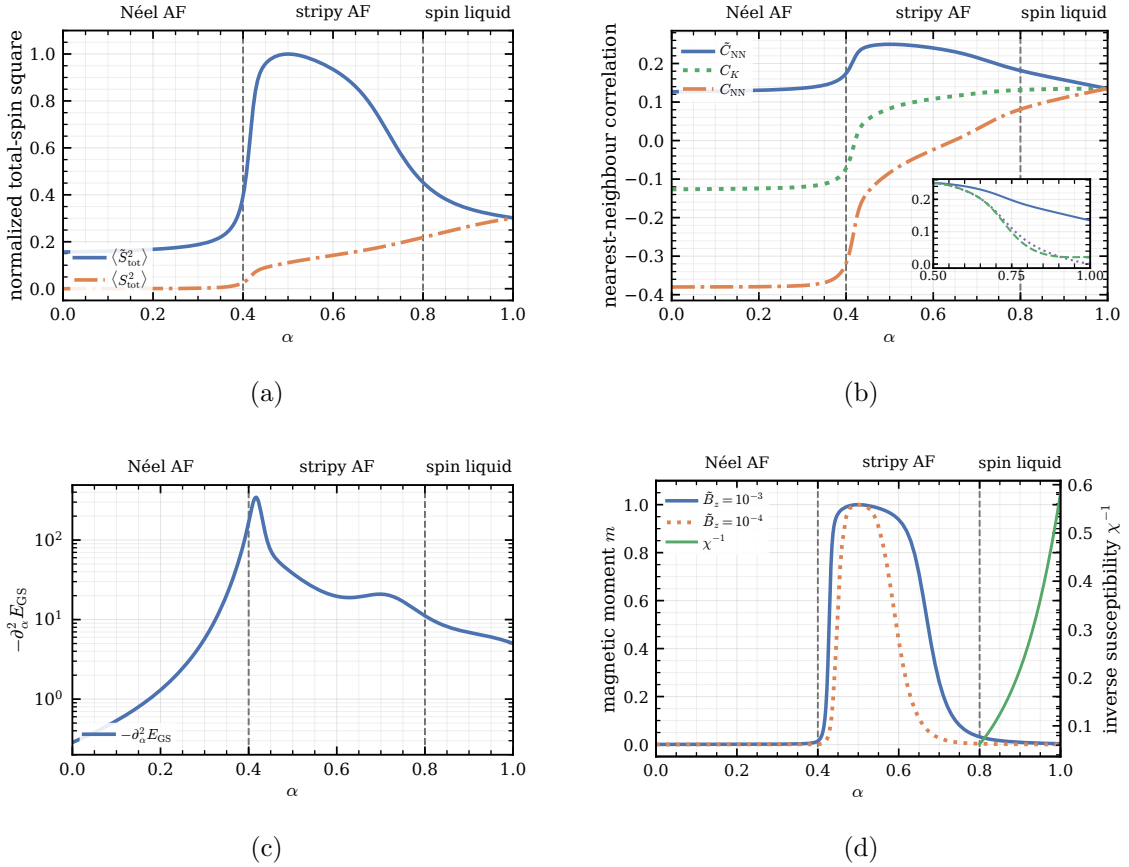



**Figure 11**: Numerical diagnostics for the finite-cluster Kitaev–Heisenberg model. (a) Normalized total-spin square in the original and rotated bases. The growth of $\langle \widetilde{\mathbf{S}}^2_{\mathrm{tot}} \rangle$ identifies the stripy phase as a hidden ferromagnet in the rotated basis. (b) Nearest-neighbour spin correlations and bond-selective Kitaev correlations. The original-basis antiferromagnetic correlations dominate near $\alpha = 0$, while rotated-basis ferromagnetic correlations dominate in the stripy region. (c) Curvature of the ground-state energy, showing rapid changes near the phase boundaries. (d) Magnetic moment in weak rotated fields and the corresponding inverse susceptibility. The dashed vertical lines mark the approximate phase boundaries between the Néel antiferromagnet, stripy antiferromagnet, and spin-liquid regimes. Plots generated numerically.

A useful way to locate where the ground state changes most rapidly is to examine the curvature of the ground-state energy,

$$-\partial_\alpha^2 E_{\mathrm{GS}}. \tag{4.45}$$

As shown in Fig. 11(c), the strongest peak appears near the Néel–stripy boundary, while a broader feature appears as the system crosses over toward the spin-liquid side. For a finite cluster these are not true singularities, but they mark the same phase-separation regions expected from the original phase diagram.

Finally, Fig. 11(d) shows the magnetic response in a weak field coupled to the rotated spins,

$$H_B(\alpha) = H(\alpha) - \widetilde{B}_z \sum_i \widetilde{S}_i^z. \tag{4.46}$$

The induced moment is

$$m = \frac{2}{N} \left\langle \sum_i \widetilde{S}_i^z \right\rangle, \tag{4.47}$$

and the inverse susceptibility is estimated from

$$\chi^{-1} \simeq \frac{\widetilde{B}_z}{m}. \tag{4.48}$$

The moment becomes large in the stripy region, as expected from the hidden ferromagnetic character of that phase. Near the Kitaev spin-liquid side, the response is suppressed and $\chi^{-1}$ rises, reflecting the weakening of conventional magnetic order.

Putting these diagnostics together, the finite-cluster calculation reproduces the expected qualitative sequence as shown in Ref. [6]:

$$\text{Néel AF} \quad \longrightarrow \quad \text{stripy AF} \quad \longrightarrow \quad \text{spin liquid.} \tag{4.49}$$

The agreement should not be understood as an exact thermodynamic phase diagram, but as a successful reproduction of the main numerical fingerprints: antiferromagnetic correlations near the Heisenberg limit, hidden ferromagnetism in the rotated basis in the stripy region, and weakened conventional magnetic response near the Kitaev spin-liquid limit.

## 5 Experimental Realizations

The Kitaev model is not only a theoretical toy model. The real question is whether a material can come close to realizing the bond-dependent exchange that makes the model so special. For this to happen, three ingredients are especially useful: localized magnetic moments, strong spin–orbit coupling, and an edge-sharing octahedral geometry. Together, these can convert ordinary magnetic exchange into highly anisotropic Kitaev-type interactions.

The basic microscopic idea is the Jackeli–Khaliullin mechanism. In spin–orbit-coupled transition-metal compounds, the spin and orbital degrees of freedom are entangled into an effective pseudospin-$\frac{1}{2}$. When the magnetic ions sit inside edge-sharing octahedra, different ligand-mediated exchange paths can interfere with one another. This interference suppresses parts of the usual Heisenberg exchange and allows bond-dependent Kitaev exchange to become important. This is why honeycomb iridates and ruthenates became the main material platforms for Kitaev physics.

The first important candidates were the layered honeycomb iridates

$$A_2\mathrm{IrO}_3, \qquad A = \mathrm{Na}, \mathrm{Li}. \tag{5.1}$$

Here the magnetic ions are $\mathrm{Ir}^{4+}$, with a $5d^5$ configuration and strong spin–orbit coupling. These materials are historically important because they motivated the Kitaev–Heisenberg model discussed in the previous section. However, real iridates contain lattice distortions, further-neighbour interactions, and additional non-Kitaev terms. They are also experimentally difficult for neutron scattering because iridium strongly absorbs neutrons. For these reasons, much of the experimental attention later shifted to $\alpha$-$\mathrm{RuCl}_3$.

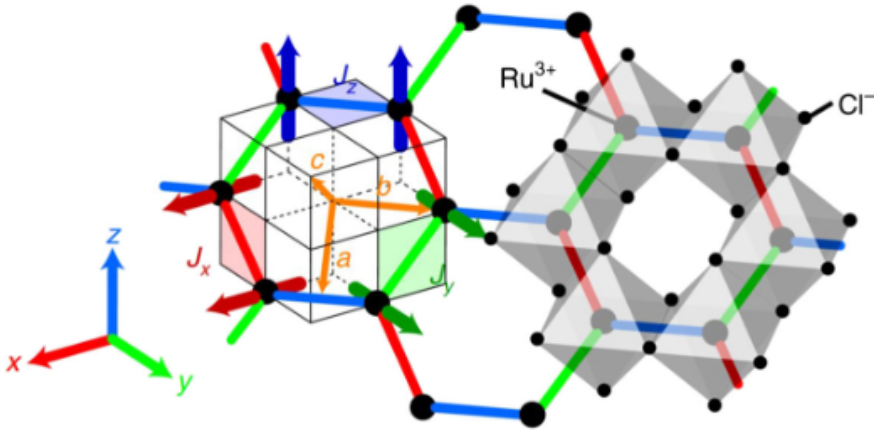



**Figure 12**: Crystal structure of $\alpha$-$\mathrm{RuCl}_3$. The Ru ions form a layered honeycomb network inside edge-sharing $\mathrm{RuCl}_6$ octahedra, making the material a natural platform for bond-dependent exchange. Figure adapted from Ref. [5].

The material $\alpha$-$\mathrm{RuCl}_3$ is now one of the most studied proximate Kitaev materials. It contains $\mathrm{Ru}^{3+}$ ions with a $4d^5$ configuration arranged on a layered honeycomb lattice. Although the spin–orbit coupling is weaker than in iridates, the combination of spin–orbit

coupling, electronic correlations, and the nearly two-dimensional honeycomb structure still produces an effective pseudospin-$\frac{1}{2}$ Mott insulating state.

But $\alpha$-$RuCl_3$ is not a perfect Kitaev spin liquid at zero field. It develops zigzag antiferromagnetic order below roughly

$$T_N \simeq 7 \text{ K}. \tag{5.2}$$

This immediately tells us that extra interactions are present. A more realistic nearest-neighbour Hamiltonian contains not only the Kitaev term, but also Heisenberg and off-diagonal anisotropic exchanges:

$$H_{ij}^{(\gamma)} = J\,\mathbf{S}_i \cdot \mathbf{S}_j + K S_i^{\gamma} S_j^{\gamma} + \Gamma \left( S_i^{\alpha} S_j^{\beta} + S_i^{\beta} S_j^{\alpha} \right) + \cdots , \tag{5.3}$$

where $\gamma = x, y, z$ labels the bond type, and $\alpha, \beta$ are the two spin components different from $\gamma$. Further-neighbour interactions, such as $J_3$, may also be important. So $\alpha$-$RuCl_3$ should be viewed as a proximate Kitaev material: close enough to show Kitaev-like fingerprints, but not close enough to avoid magnetic order completely.

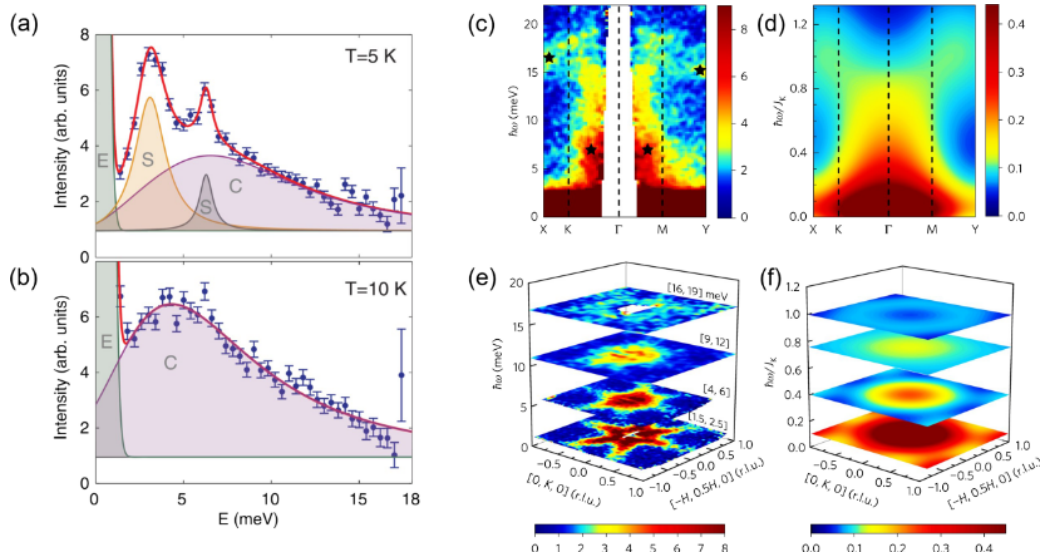



**Figure 13**: Inelastic neutron scattering response of $\alpha$-$RuCl_3$. Panels (a) and (b) show energy-dependent spectra below and above the zigzag ordering temperature, respectively: sharp spin-wave modes appear in the ordered phase, while a broad magnetic continuum remains visible above $T_N$. Panel (c) shows the measured momentum- and energy-resolved neutron scattering intensity, with a broad continuum extending over a wide energy range. Panel (d) shows the corresponding theoretical response expected from the Kitaev model, capturing the broad continuum-like structure. Panels (e) and (f) compare constant-energy cuts from experiment and theory, showing that several qualitative features of the scattering pattern are reproduced by the Kitaev-type description. The persistence of a broad continuum is one of the main experimental motivations for viewing $\alpha$-$RuCl_3$ as a proximate Kitaev spin-liquid material. Figure adapted from Ref. [5].

The experimental evidence for proximity to Kitaev physics comes from several probes. Specific heat shows broad features that have been interpreted as a two-stage entropy release associated with Majorana fermions and flux excitations. Raman and terahertz spectroscopy observe broad magnetic continua that are difficult to explain using only sharp magnon modes. Inelastic neutron scattering also reveals a broad continuum over a large energy range, especially above the ordering temperature. This continuum is one of the main reasons $\alpha$-$RuCl_3$ is discussed as a proximate spin liquid rather than as an ordinary magnet.

A further route is magnetic-field tuning. An in-plane magnetic field of order

$$\mu_0 H_c \sim 7\text{–}8 \text{ T} \tag{5.4}$$

suppresses the zigzag order in $\alpha$-$RuCl_3$, producing a field-induced disordered regime. Whether this regime is a true Kitaev-like quantum spin liquid, a partially polarized paramagnet, or something more complicated remains debated. Thermal transport is especially interesting here because Majorana fermions are charge neutral but can carry heat. Thermal Hall measurements have therefore attracted attention, although their interpretation is still subtle because phonons, disorder, sample dependence, and non-Kitaev interactions can also affect the signal.

The main lesson is simple. Real materials do not realize the pure Kitaev Hamiltonian exactly. They realize extended Kitaev models, with Heisenberg, $\Gamma$, $\Gamma'$, and further-neighbour interactions. This is precisely why the Kitaev–Heisenberg model is useful: it is the simplest bridge between the exactly solvable Kitaev spin liquid and the ordered magnetic phases seen in experiments.

## 6 Conclusion

In this report, we studied parton mean-field theory as a language for quantum spin liquids. The basic idea is to rewrite spins using auxiliary fractionalized particles and describe their hopping, pairing, and gauge structure at mean-field level. This enlarges the Hilbert space, so constraints and gauge redundancies must be treated carefully, but it gives a natural framework for phases where the usual picture of ordered spins and magnons is insufficient.

The Heisenberg model served as the simplest example, showing how spin exchange becomes a quartic spinon interaction that can be decoupled into hopping and pairing channels. The Kitaev model then provided the clean benchmark: using Majorana partons, spins fractionalize exactly into itinerant Majorana fermions and $\mathbb{Z}_2$ gauge fluxes, reducing the problem to free Majorana hopping in a fixed flux sector.

We then considered the Kitaev–Heisenberg model, where the Kitaev spin liquid competes with ordinary Heisenberg exchange. The four-sublattice rotation explains the intermediate stripy antiferromagnet as a hidden ferromagnet in the rotated basis, while the numerical calculation reproduces the main finite-cluster signatures of the phase diagram. Finally, real materials such as $\alpha$-$RuCl_3$ show why this problem matters experimentally: although they contain interactions beyond the pure Kitaev model, broad continua in neutron scattering and related probes suggest proximity to Kitaev spin-liquid physics.

Overall, parton mean-field theory is not just a calculational approximation, but a different way of organizing the problem: the Heisenberg model teaches the method, the Kitaev model shows exact fractionalization, and the Kitaev–Heisenberg model shows how fractionalized physics competes with conventional magnetic order.

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